BONES OF THE UPPER LIMB

Department of Anatomy

Musculoskeletal System

The Upper Limb (1)

Second Part

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Lecture Points

- **Skeletal System Parts.**
- **Structural Types of Bones**
- JOINTS
  - Definition.
  - Type of joints.
  - Typical synovial joint .
  - The accessory structures associated with synovial joint and major features.
  - Types of synovial joint according the shape.
  - Type of Cartilaginous joints:
- Skelton of Musculoskeletal system.
- The Upper Limbs
- *The Bones of Upper Limb are:*
- *Pectoral or Shoulder Girdle*
- Clavicle
  - Articulation of the clavicle
  - Muscle Origins and Insertions and Ligament attachments
  - Normal X-ray
  - Fractures of the Clavicle
- Scapula.
  - Fracture of the Scapula
  - Articulation of head of humorous with scapula
  - Humerus
    - Fractures of Surgical Neck and Mid shaft Humerus
    - Supracondylar Fractures.
    - Nerves affected in Fractures of Humerus
- Bones of The Forearm
  - Ulna
  - Radius
  - Normal X-ray Radiographic Appearances
  - Fractures of radius & ulna
  - fractured elbow
- WRIST (CARPUS)
- Carpal Bones
- Metacarpus
- Digits (Phalanges).

Scapula

- It is a **triangular.**
- It is **Flat bone.**
- **Extends between the 2nd - 7th ribs.**

It has :

- **Two Surfaces:**

1) Concave Anterior (Costal):

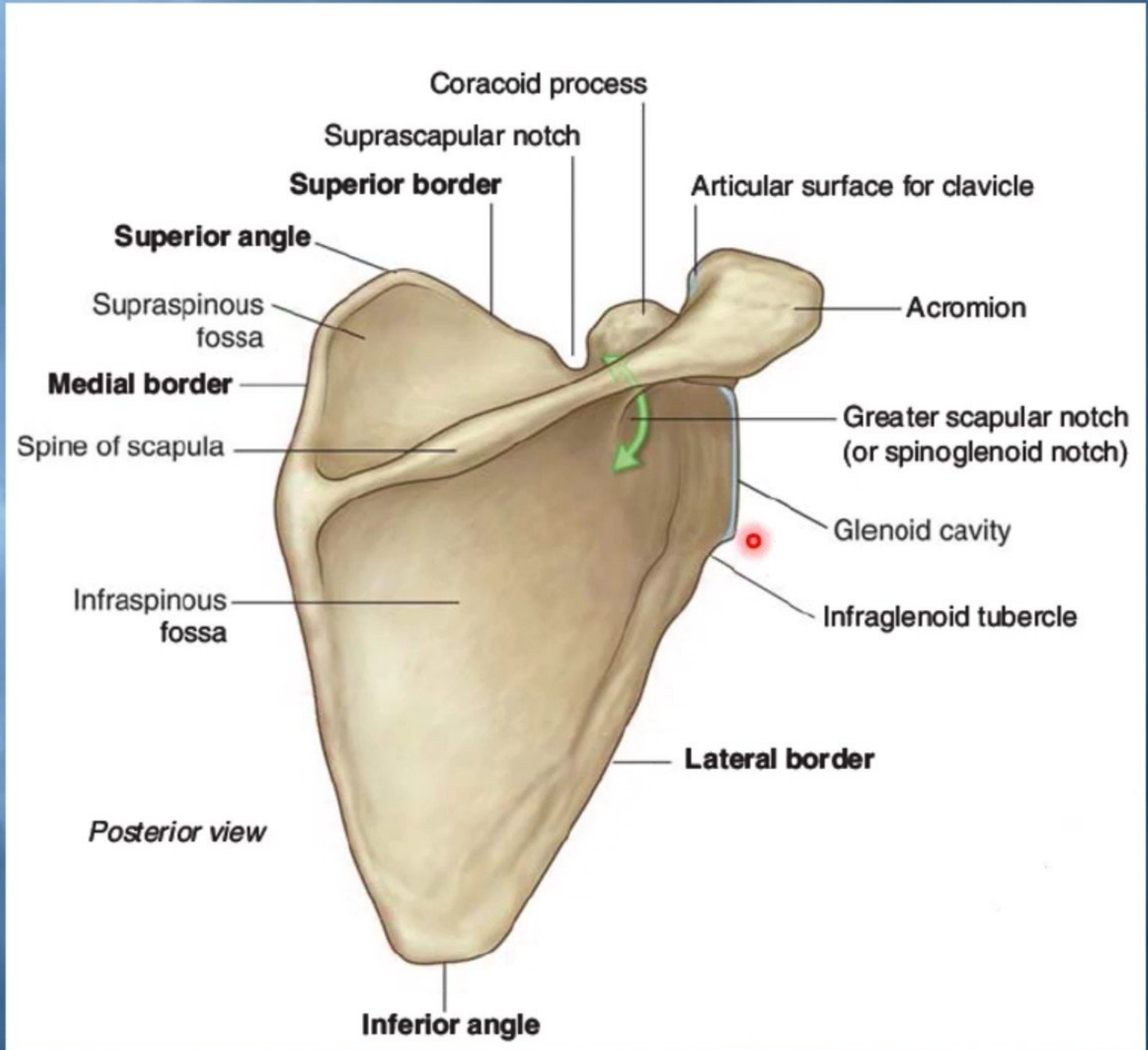
It forms the **large Subscapular Fossa**.

2) Convex Posterior :

Divided by the spine of the scapula into the:

- **Smaller Supraspinous Fossa (above the spine).**
- **larger Infraspinous Fossa (below the spine).**

- **Suprascapular notch**



Scapula

Three Angles :

- ***Superior***
- ***Lateral** (forms the Glenoid cavity) : a shallow concave oval fossa that receives the head of the humerus.*
- ***Inferior***

Three Processes:

(1) Spine: *a thick projecting ridge of bone that continues laterally.*

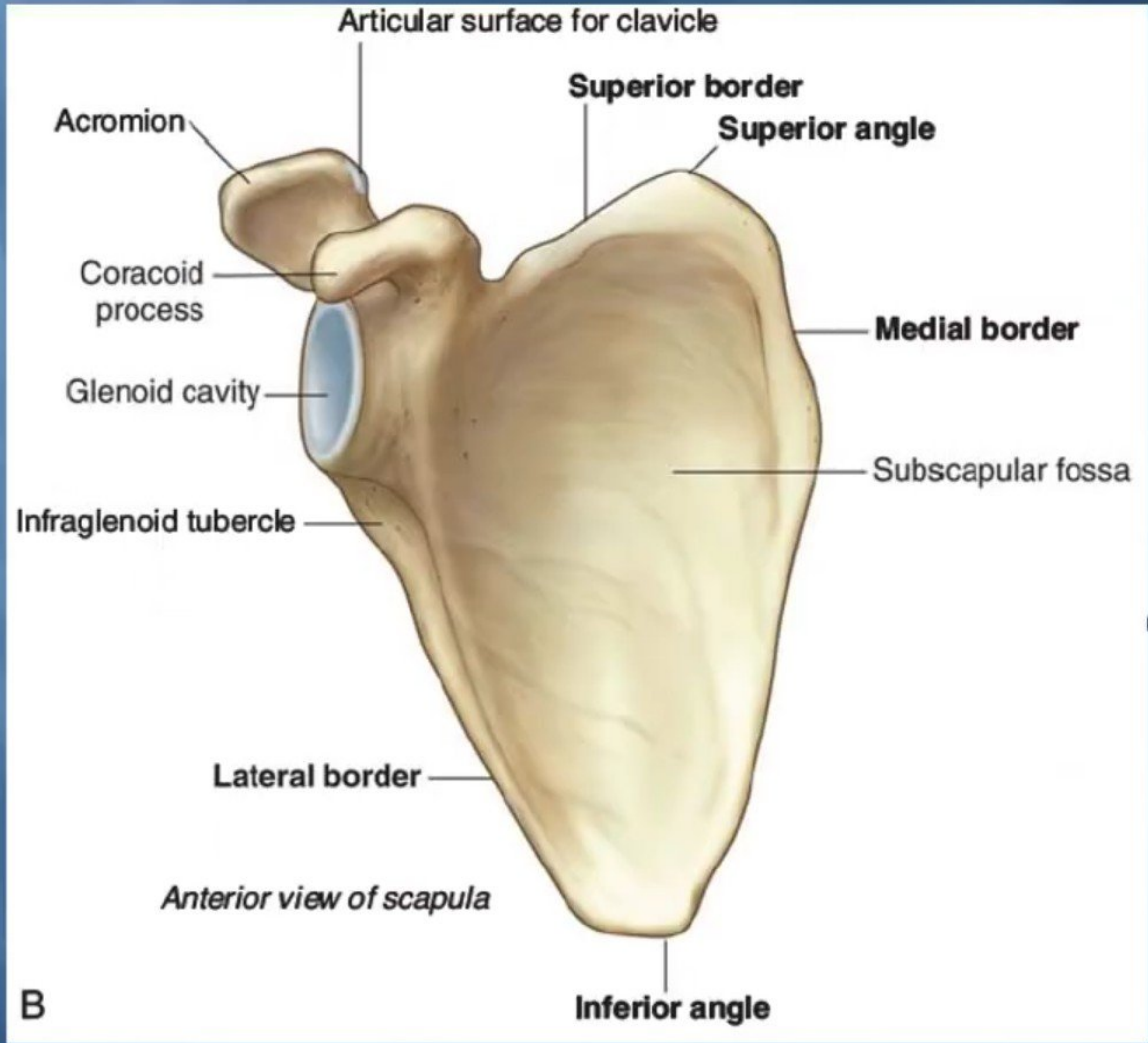
(2) Acromion *: forms the subcutaneous point of the shoulder.*

(3) Coracoid: *a beak-like process.*

- *It resembles in size, shape and direction a bent finger pointing to the shoulder.*

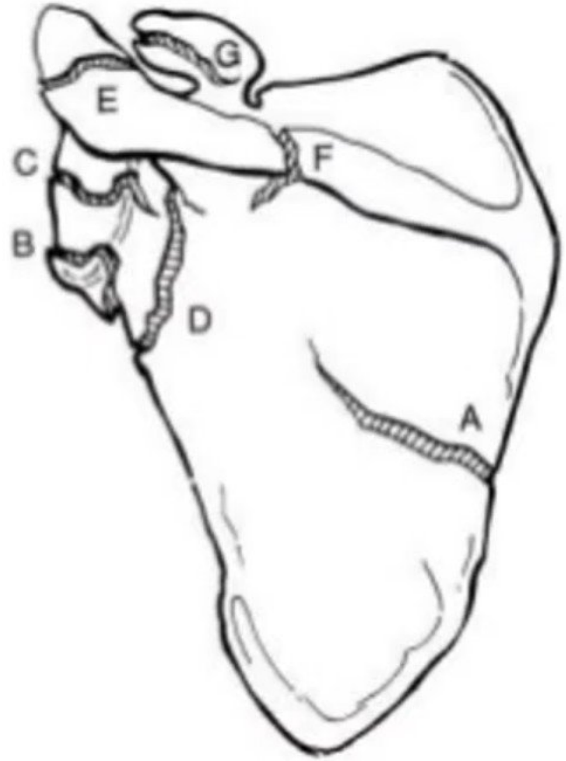
Three Borders:

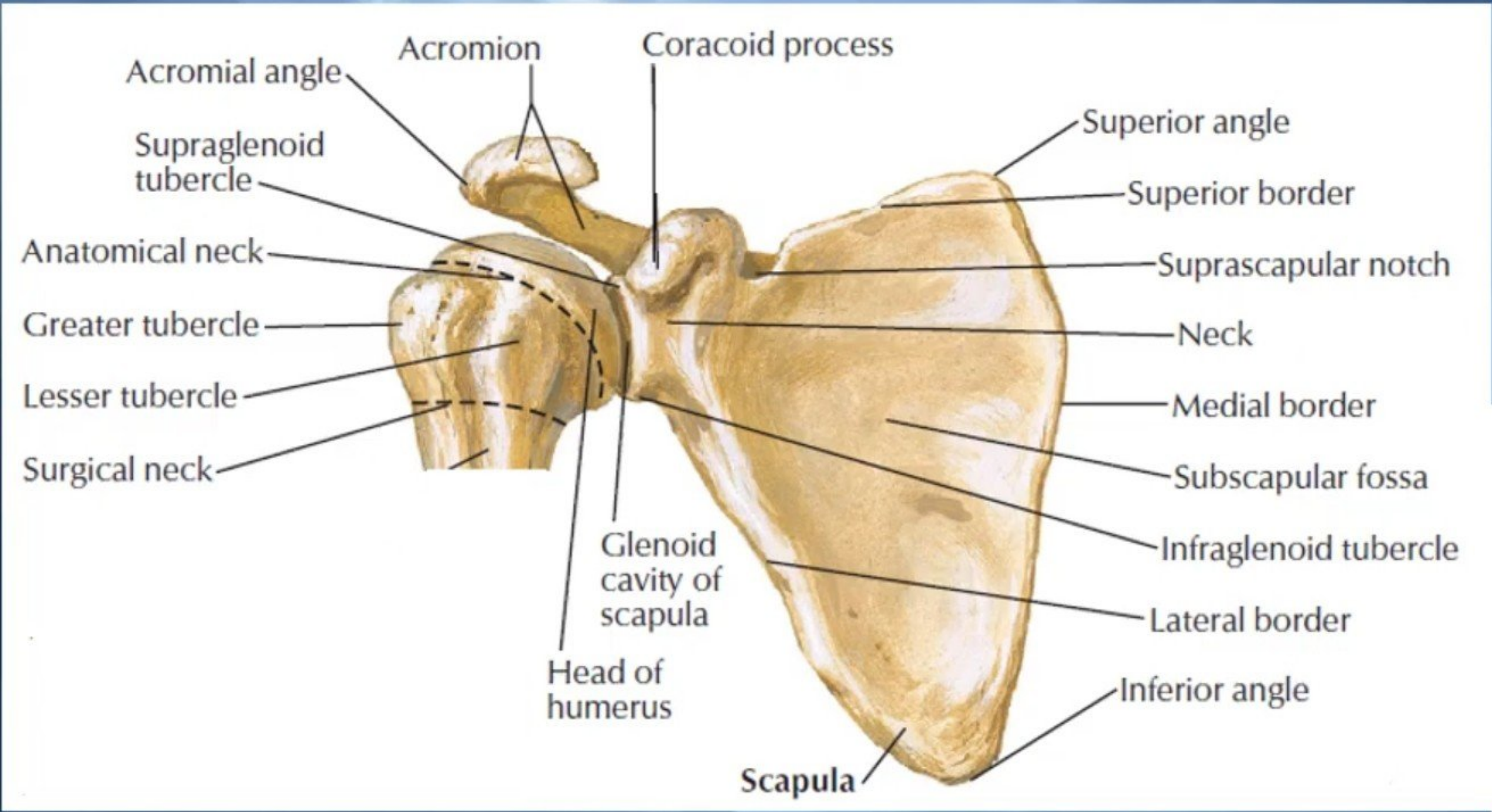
- ***Superior, Medial (Vertebral) & Lateral (Axillary)** (the thickest) part of the bone, it terminates at the lateral angle .*

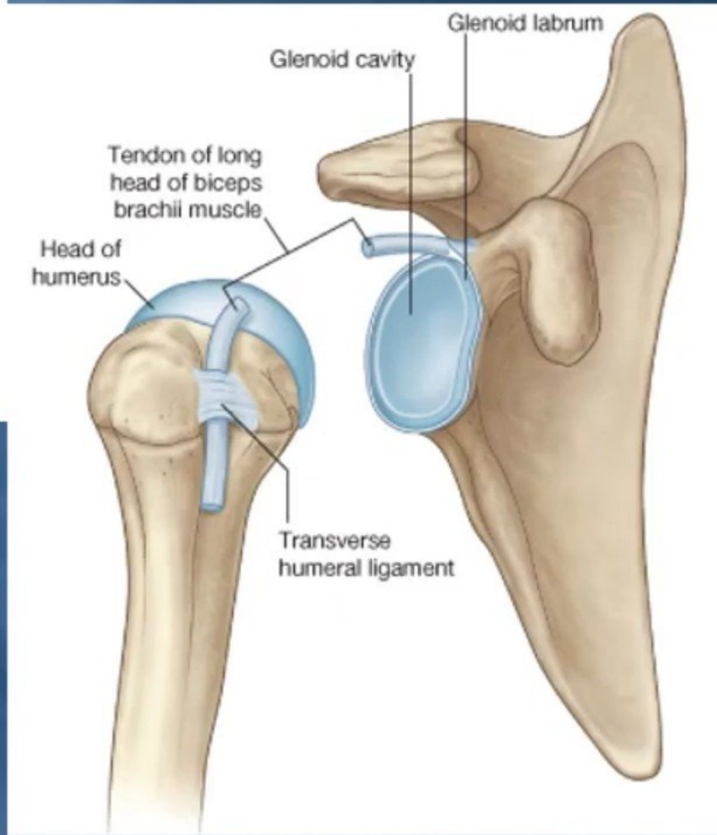


Fracture of the Scapula

- **Definition:** Is a fracture of shoulder blade, represent an uncommon injury.
- **Types:**
  - Body (A).
  - Neck (D).
    - Type I - nonangulated, nondisplaced
    - Type IIa - shortened / displaced > 1 cm.
    - Type IIb - Angulated > 40 degree.
  - Glenoid (B C).
  - Acromian (E).
  - Coracoid (G).







Articulation of head of humorous with scapula

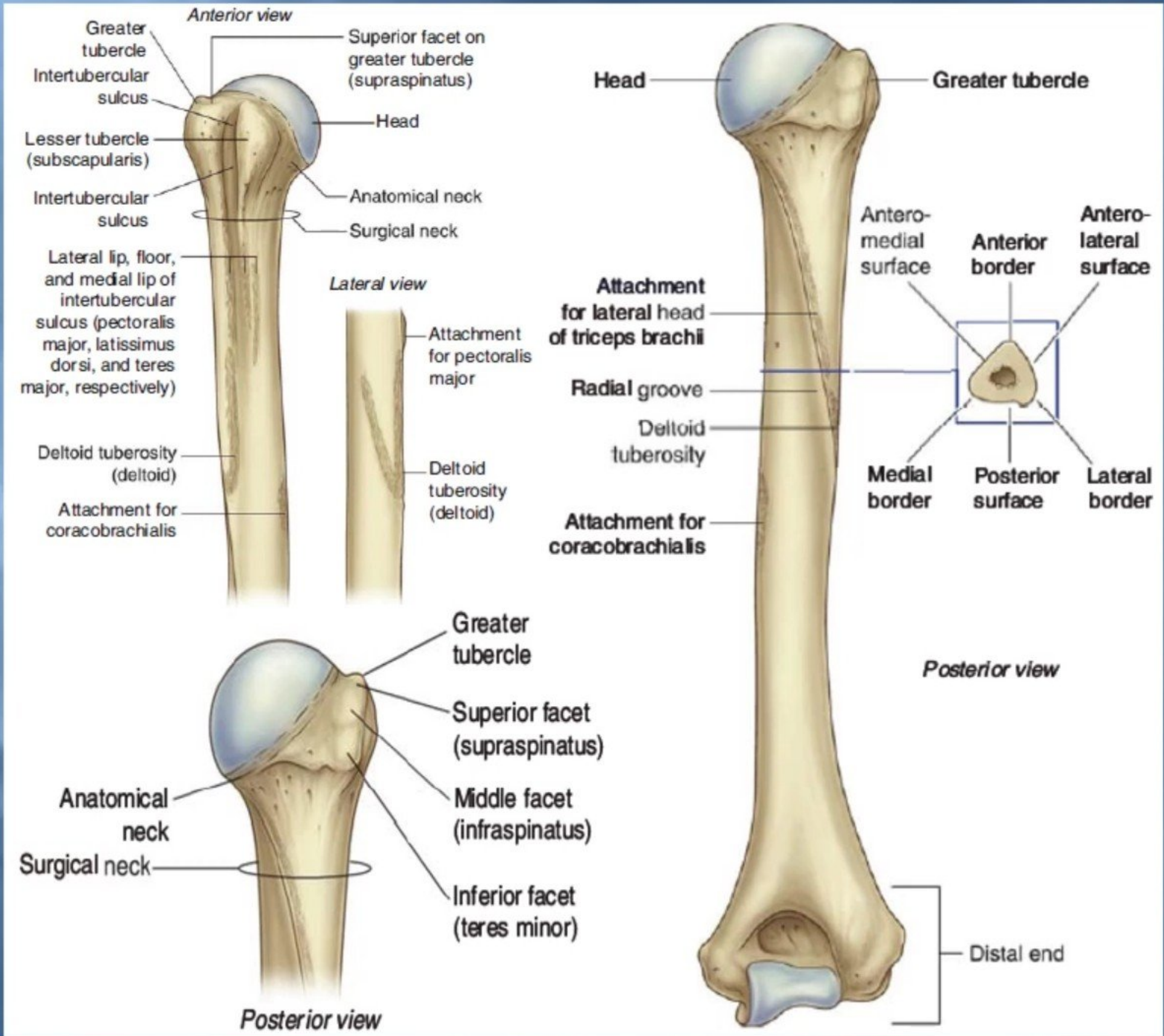
Humerus

A typical ***Long bone.***

- ***It is the largest bone in the Upper Limb***

Proximal End :

- ***Head, Neck, Greater & Lesser Tubercles.***
- ***Head:*** ***Smooth***
- ***It forms 1/3 of a sphere, it articulates with the glenoid cavity of the scapula.***
- ***Greater tubercle:*** ***at the lateral margin of the humerus.***
- ***Lesser tubercle:*** ***projects anteriorly.***
- ***The two tubercles are separated by Intertubercular Groove.***
- ***Anatomical neck:*** ***formed by a groove separating the head from the tubercles***
- ***Surgical Neck:*** ***a narrow part distal to the tubercles.***



Humerus

Shaft (Body):

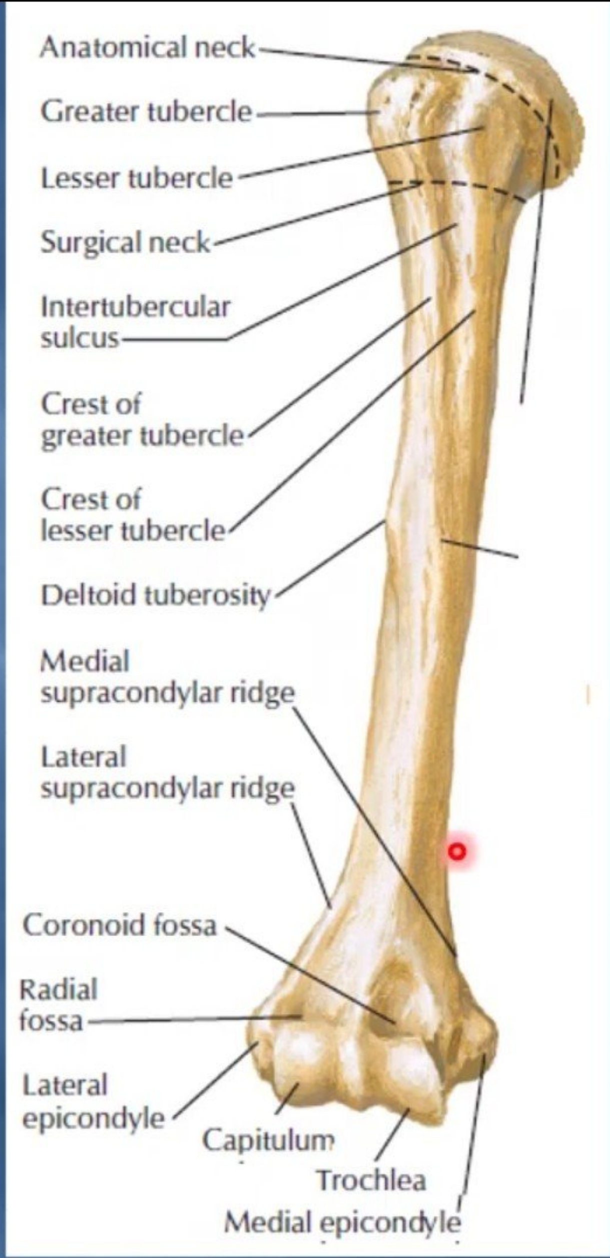
- ***Has two prominent features:***

1) Deltoid tuberosity:

- ***A rough elevation laterally for the attachment of deltoid muscle.***

2) Spiral (Radial) groove:

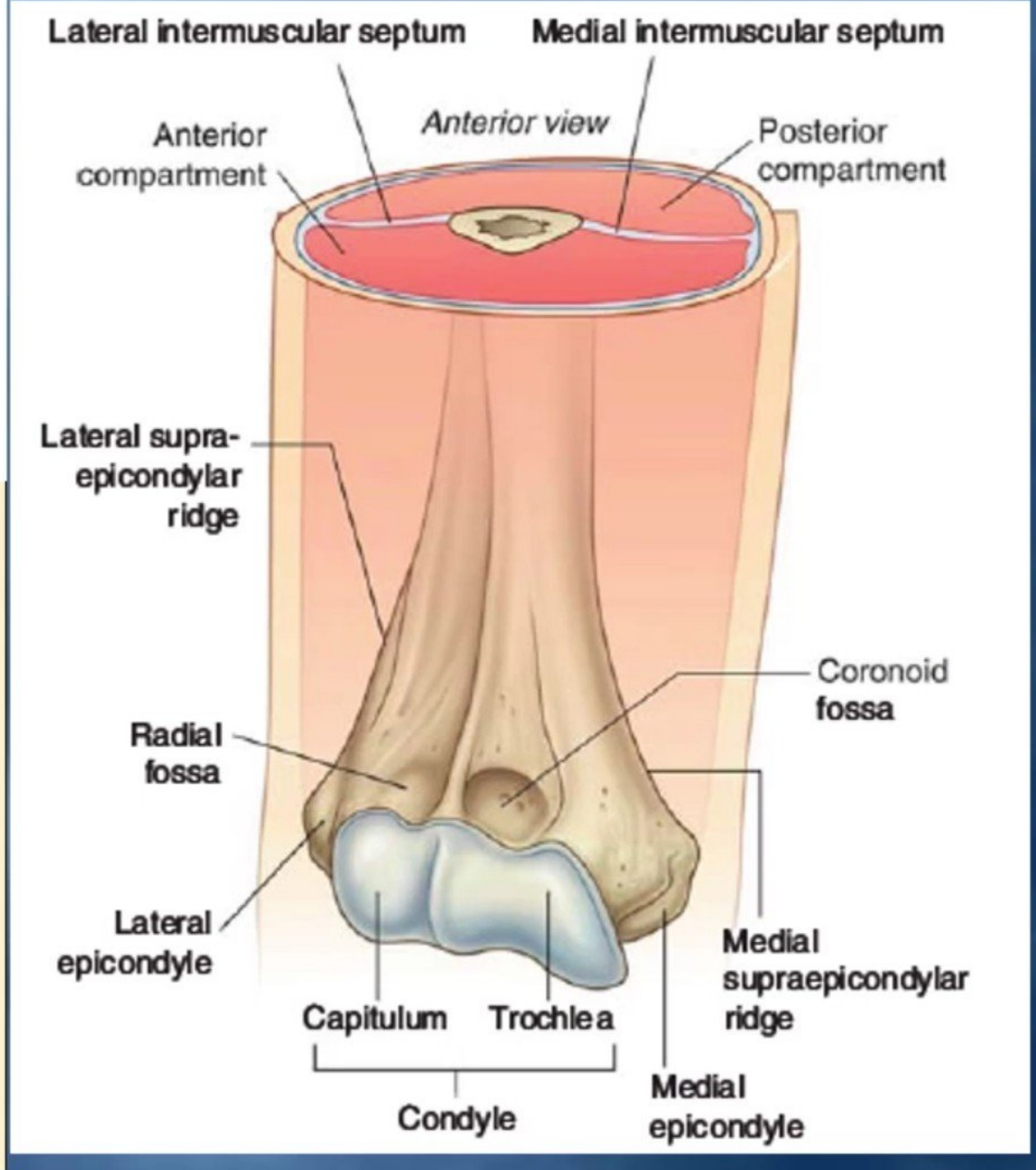
- ***Runs obliquely down the posterior aspect of the shaft.***
- ***It lodges the important radial nerve & profunda brachii vessels.***
- ***Distal End:***
- ***Widens as the sharp medial and lateral Supracondylar Ridges and end in the Medial (can be felt) and Lateral Epicondyles.***
- ***They provide muscular attachment.***



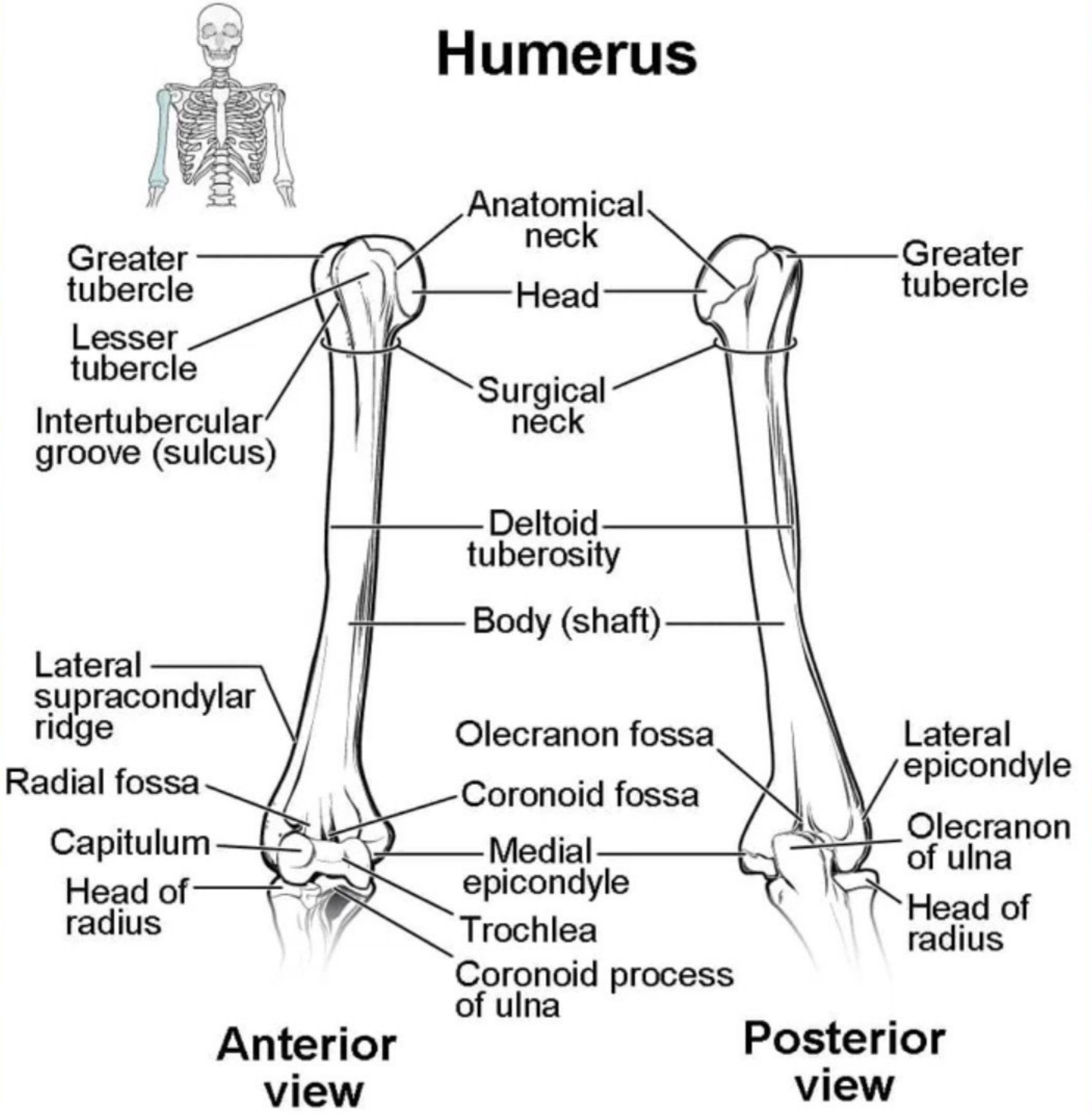
Humerus

Distal end:

- **Anteriorly:**
- **Trochlea:** (medial) for articulation with the ulna
- **Capitulum:** (lateral) for articulation with the radius.
- **Coronoid fossa :** above the trochlea.
- **Radial fossa:** above the capitulum.
- **Posteriorly:**
- **Olecranon fossa :** above the trochlea.

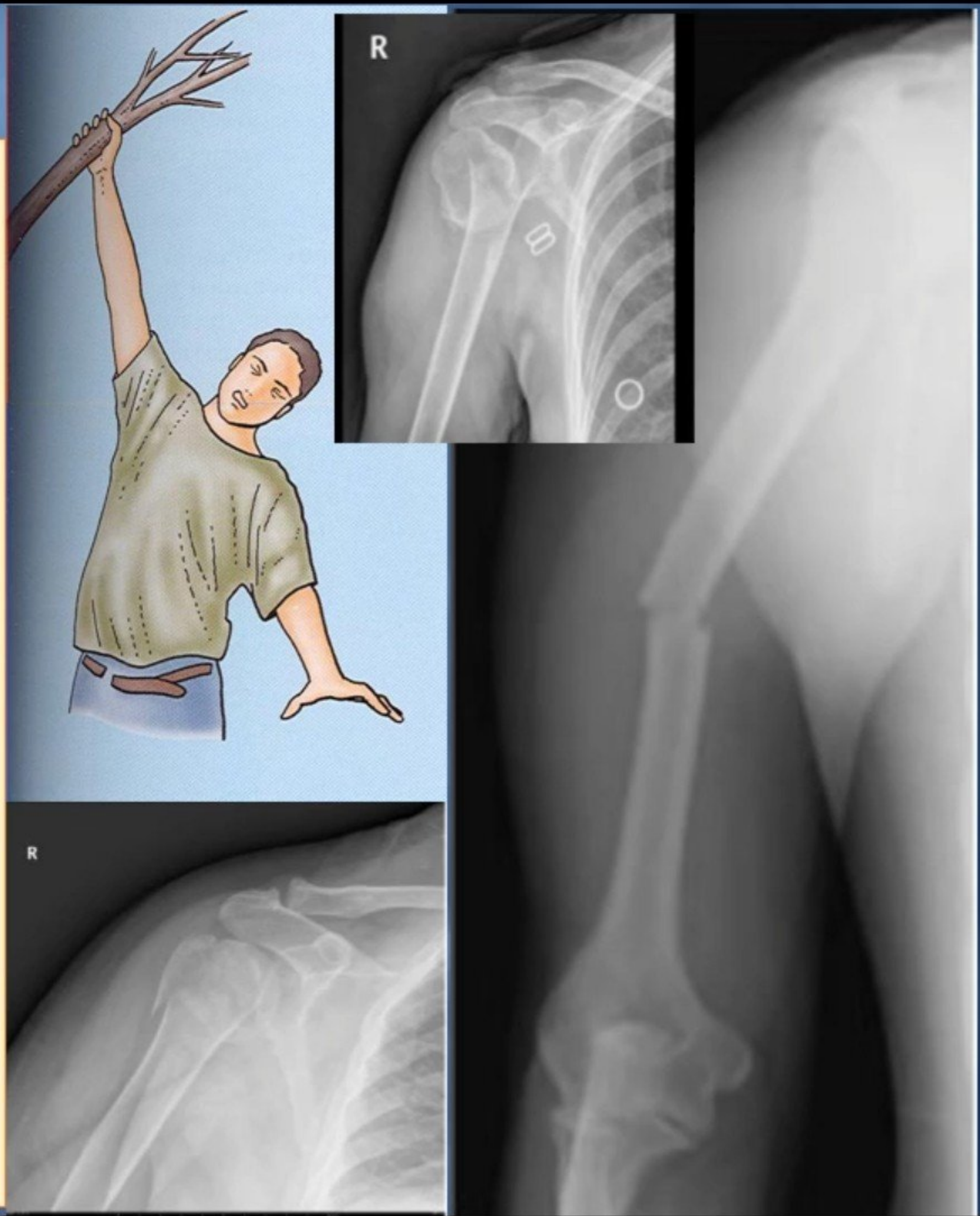


Humerus


Anatomical neck
Greater tubercle
Head
Greater tubercle
Lesser tubercle
Surgical neck
Intertubercular groove (sulcus)
Deltoid tuberosity
Body (shaft)
Lateral supracondylar ridge
Olecranon fossa
Lateral epicondyle
Radial fossa
Coronoid fossa
Capitulum
Medial epicondyle
Olecranon of ulna
Head of radius
Trochlea
Head of radius
Coronoid process of ulna
Anterior view
Posterior view

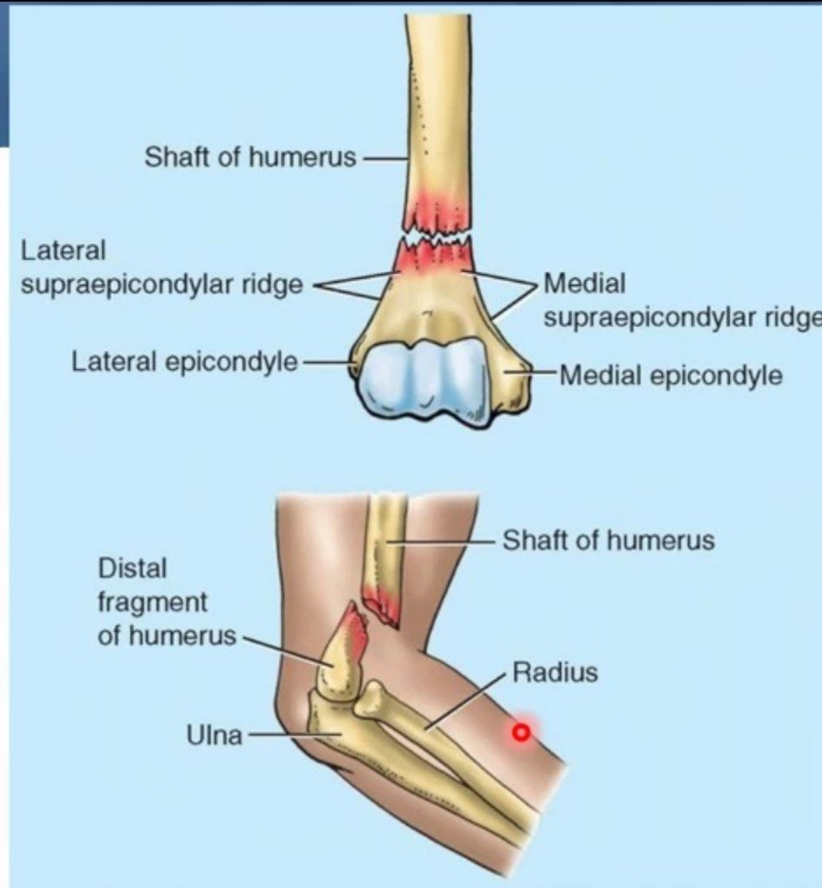
Fractures of Humerus

- Most common fractures are of the **Surgical Neck** especially in **elder people with osteoporosis.**
- The fracture results from **falling on** the hand (transition of force through the bones of forearm of the extended limb).
- In **younger people**, fractures of the **greater tubercle** results from falling on the hand when the arm is abducted .
- The **shaft of the humerus (mid shaft)** can be fractured by a direct blow to the arm or by indirect injury as falling on the outstretched hand.



Supracondylar Fractures

- A **fracture of the distal part** of the humerus, near the supraepicondylar ridges, is called **a supraepicondylar fracture** .
- The distal bone fragment may be **displaced anteriorly** or **posteriorly.**
- The actions of the **brachialis** and **triceps** tend to pull the distal fragment over the proximal fragment, **shortening the limb**. Any of the nerves or branches of the brachial vessels related to the humerus may be injured by a displaced bone fragment



Nerves Affected in Fractures of Humerus

Surgical neck:

Axillary nerve

Musculocuanous nerve

Not related to bone

Distal end of humerus:

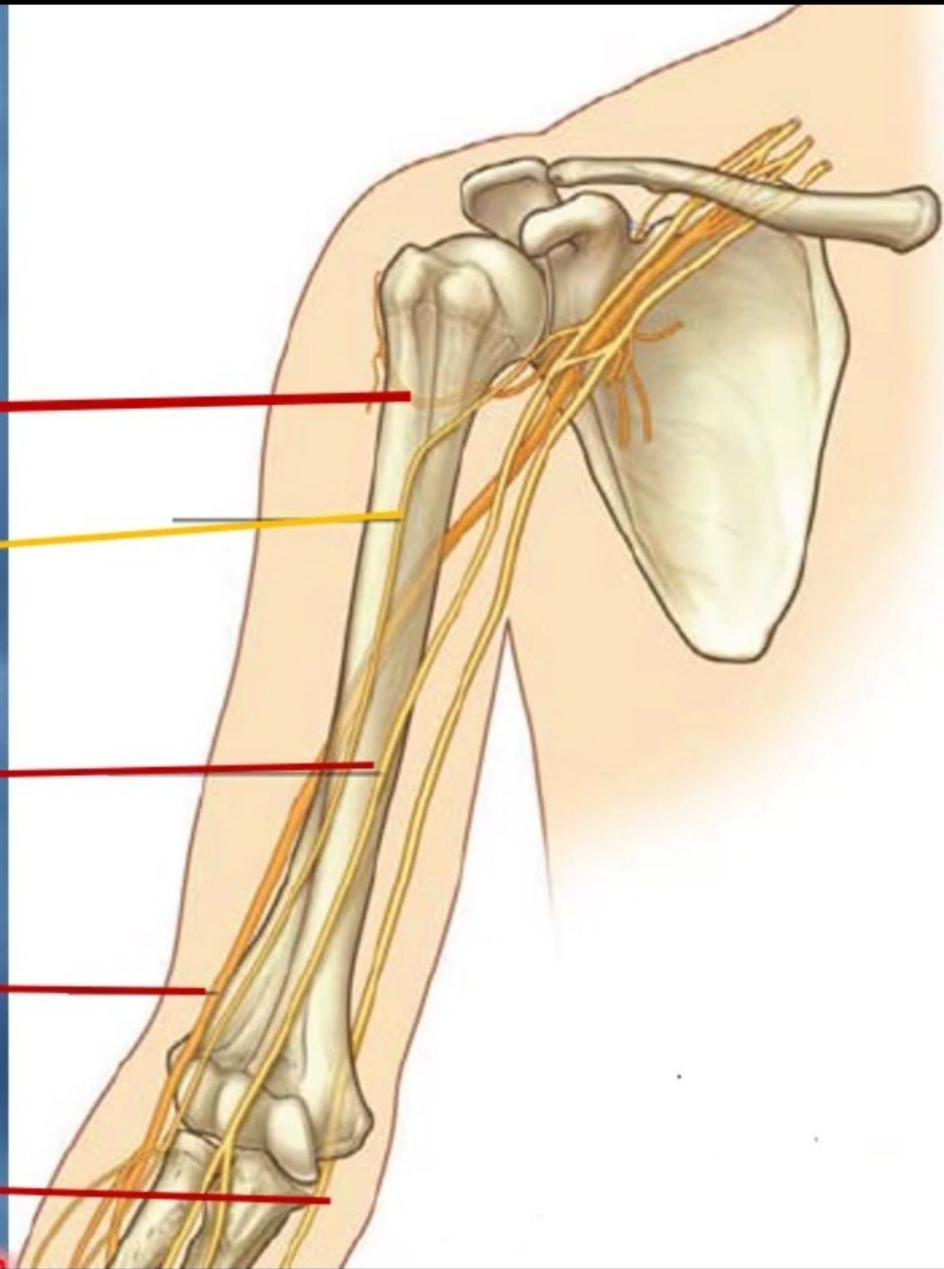
Median nerve.

Radial groove (shaft):

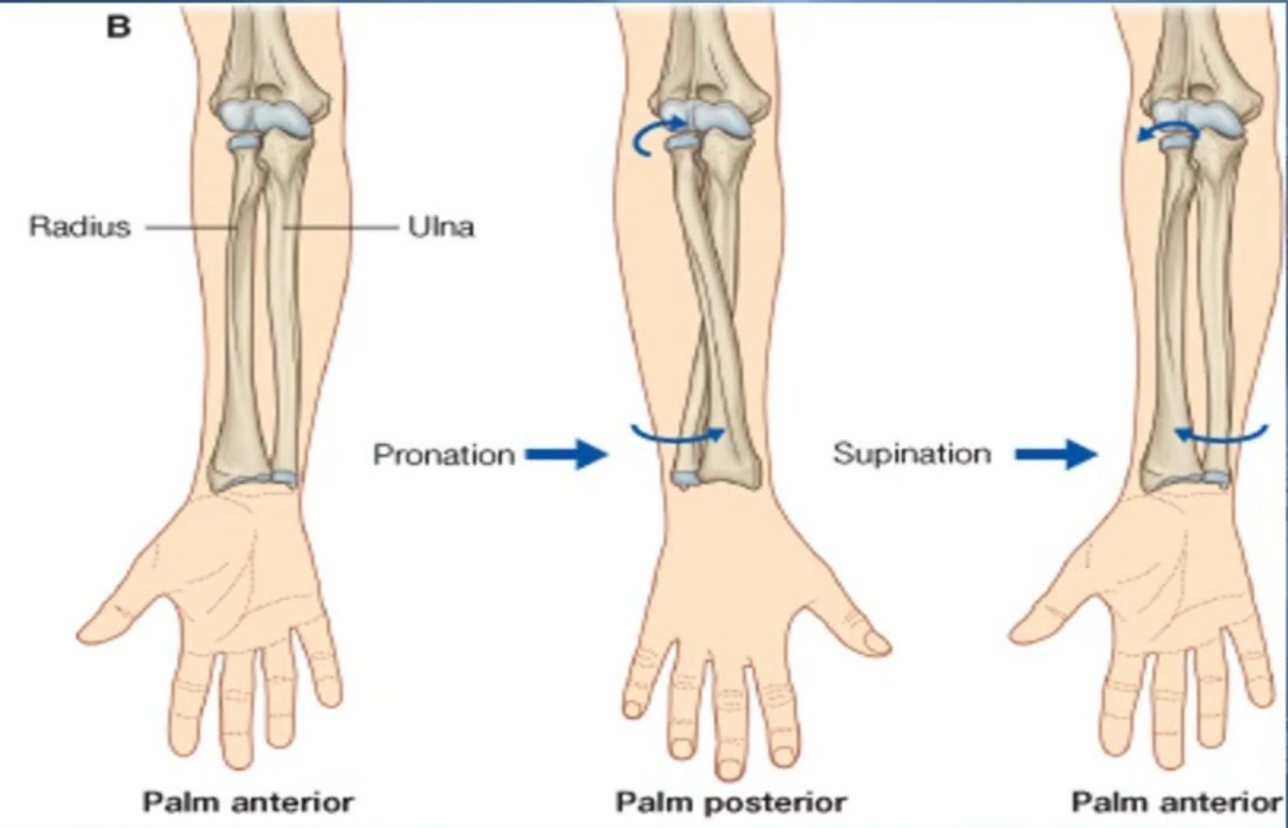
Radial nerve.

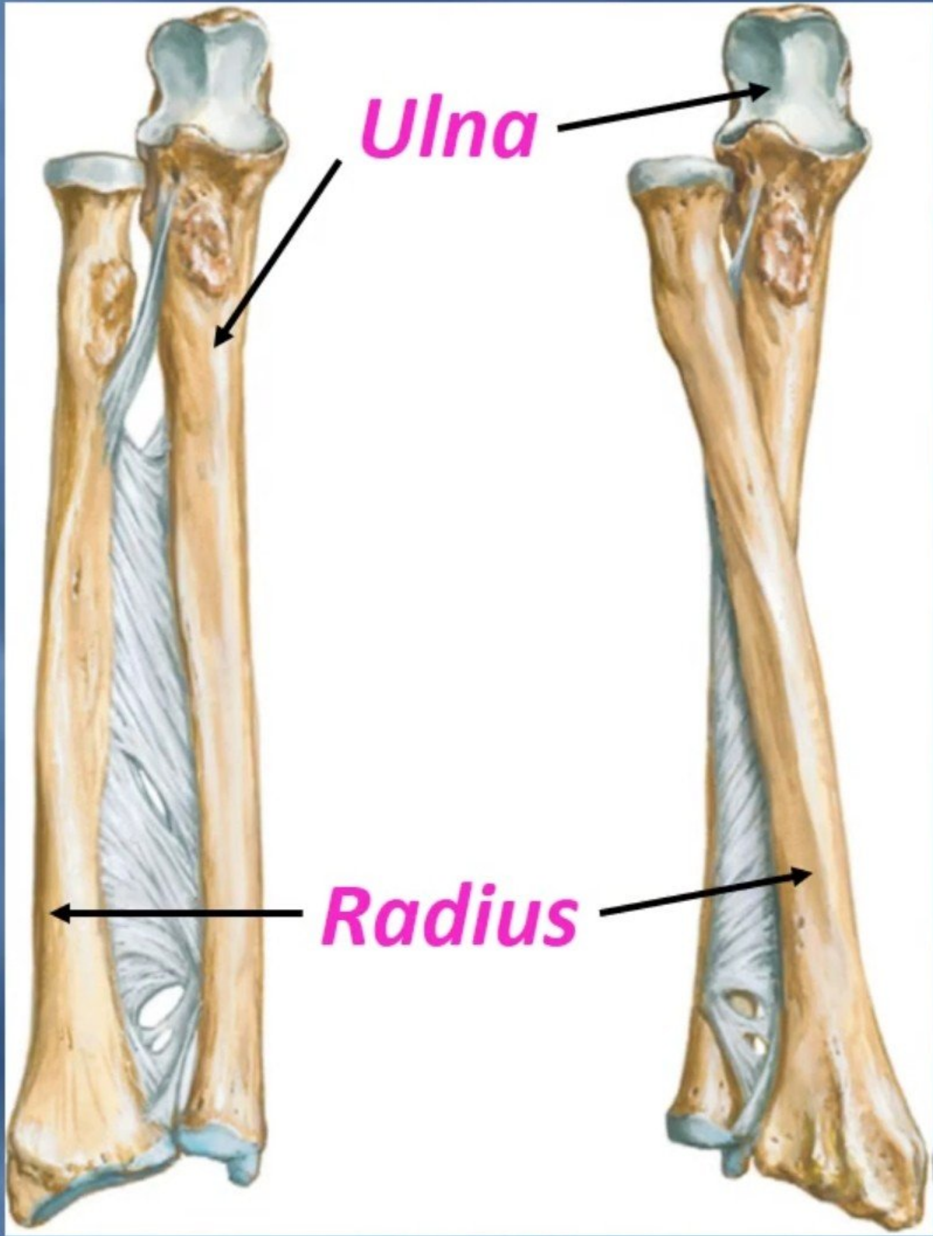
Medial epicondyle :

Ulnar nerve.



Bones of The Forearm



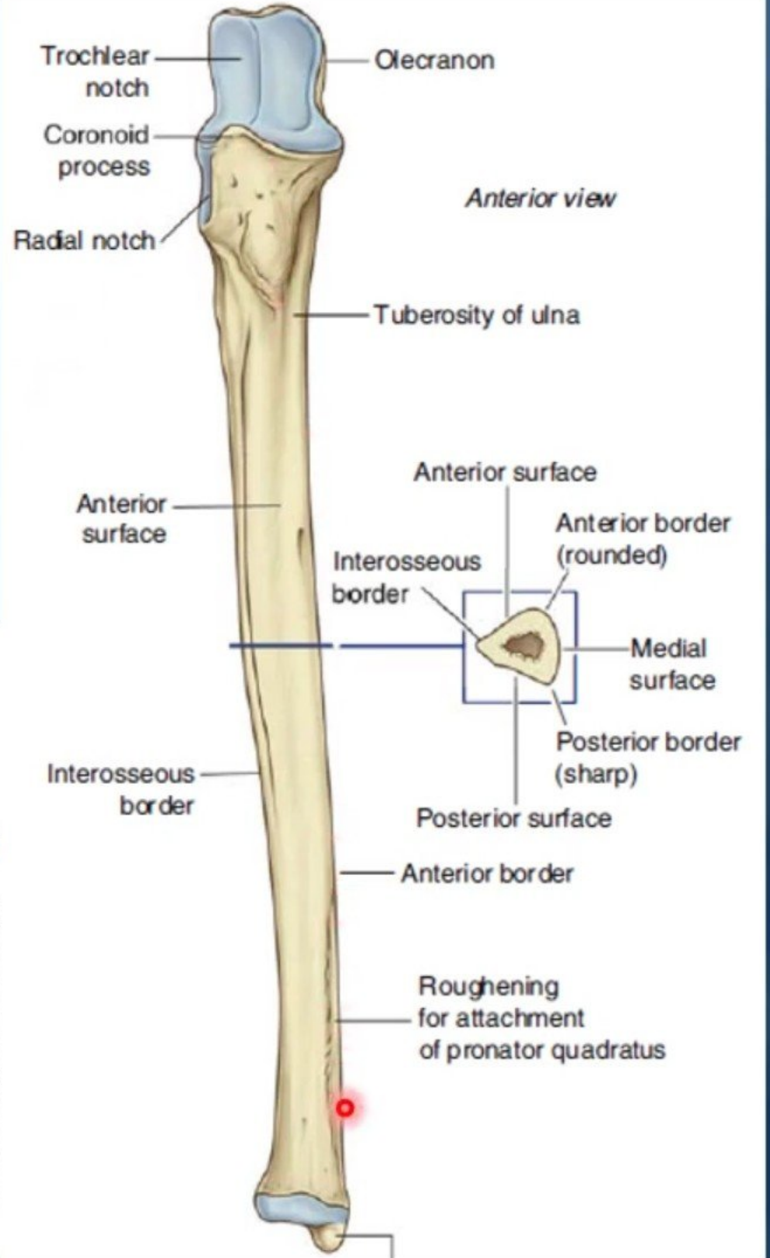


Ulna

- Olecranon
- Coronoid Process
- Trochlear Notch
- Radial Notch

Lateral:

Anterior:



- *It is the stabilizing bone of the forearm.*
- *It is the medial & longer of the two bones of the forearm.*

Proximal End

1. Olecranon Process :

- *projects proximally from the posterior aspect(forms the prominence of the elbow).*

2. Coronoid Process :

- *projects anteriorly.*

3.Tuberosity of Ulna:

- *inferior to coronoid process.*

4.Trochlear Notch:

- *articulates with trochlea of humerus.*

5.Radial Notch :

- *a smooth rounded concavity lateral to coronoid process.*

Ulna

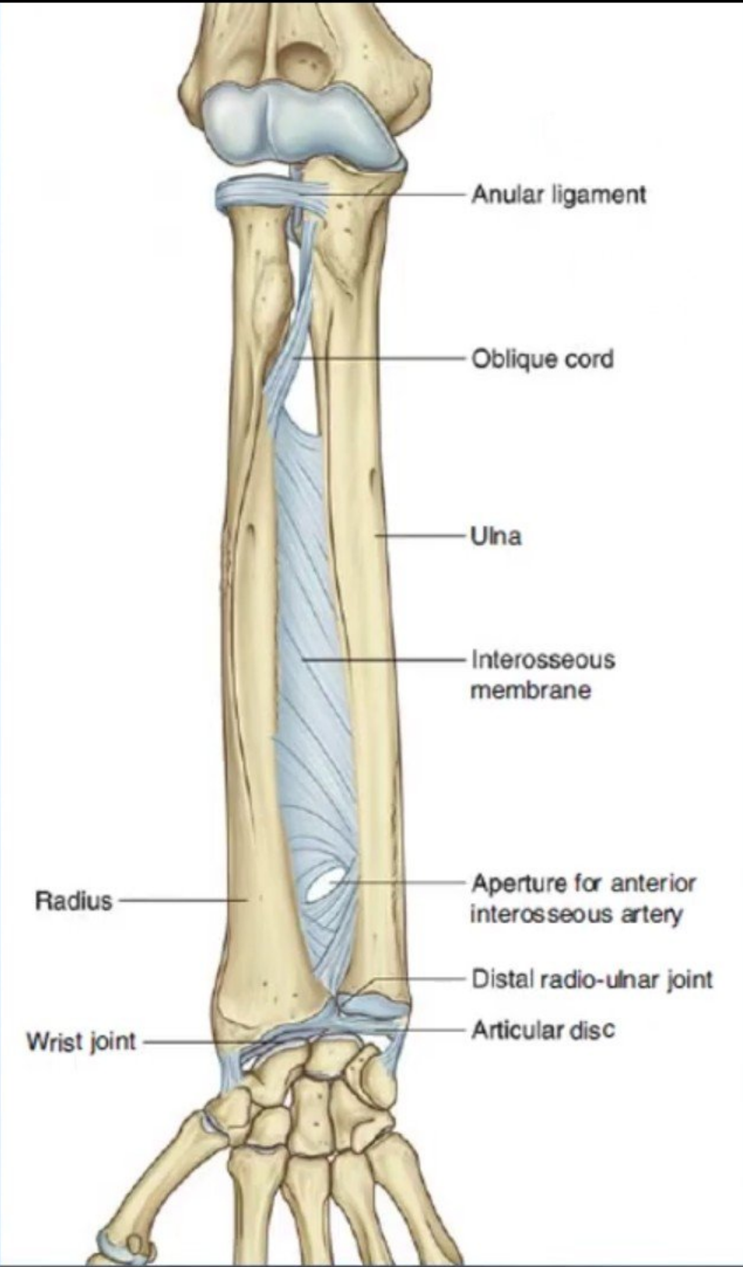
Shaft :

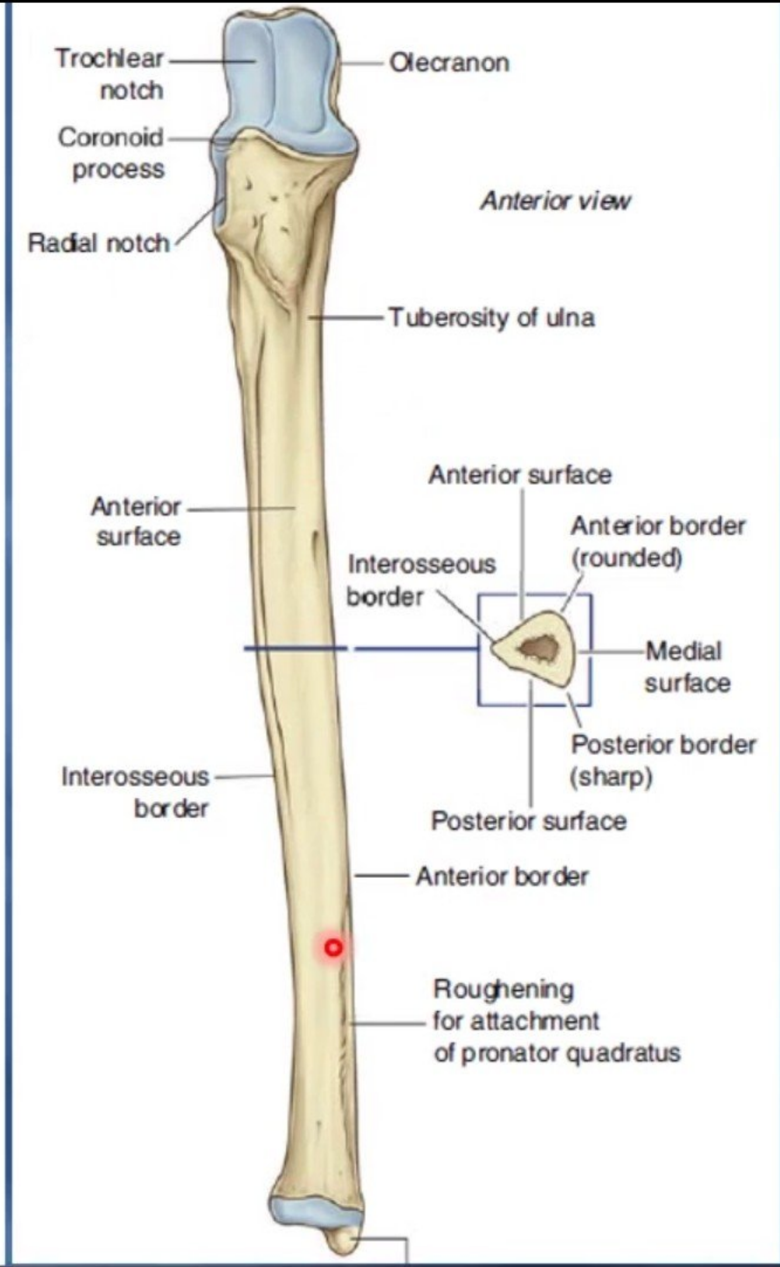
- *Thick & cylindrical superiorly but diminishes in diameter inferiorly*
- *It has Three Surfaces (Anterior, Medial & Posterior).*
- *Sharp Lateral Interosseous border.*

Distal End:

- *Small rounded*

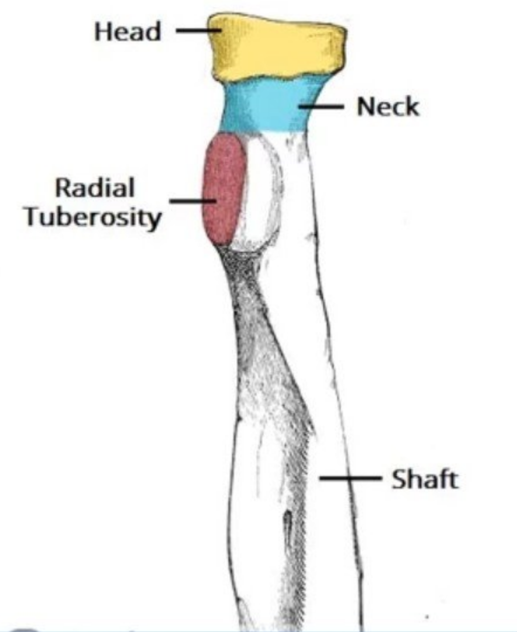
1. *Head: lies distally at the wrist. .*
2. *Styloid process: Medial.*

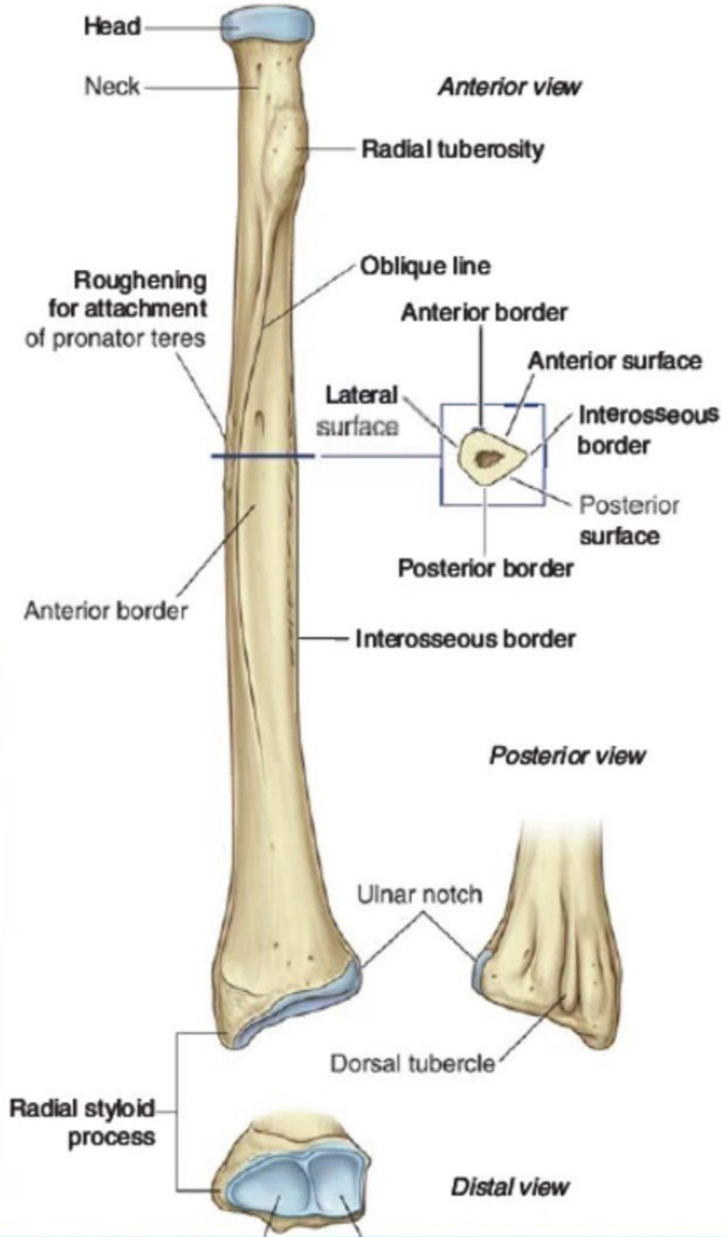




Radius

- *It is the shorter and lateral of the two forearm bones.*
- **Proximal End:**
- **1. Head**: small & circular
- **Its upper surface is concave for articulation with the Capitulum.**
- **2. Neck.**
- **3. Radial (Biciptal) Tuberosity : medially directed and separates the proximal end from the body.**
- **Shaft:**
- *Has a lateral convexity.*
- *It gradually enlarges as it passes distally.*





Radius

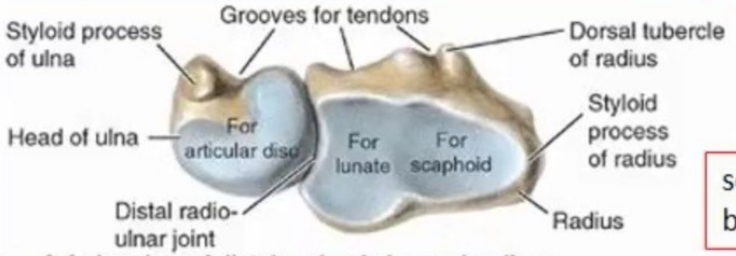
Distal (Lower) End:

It is rectangular

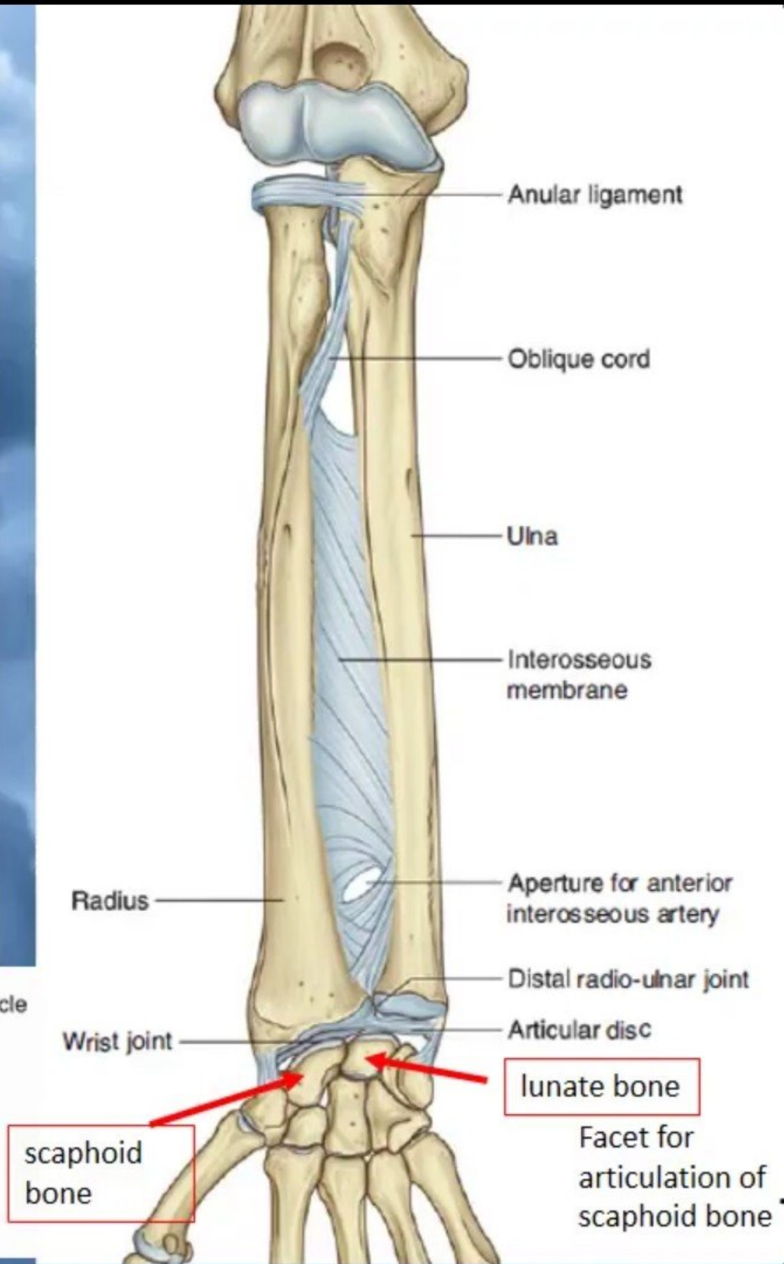
1. Ulnar Notch : a medial concavity to accommodate the head of the ulna.

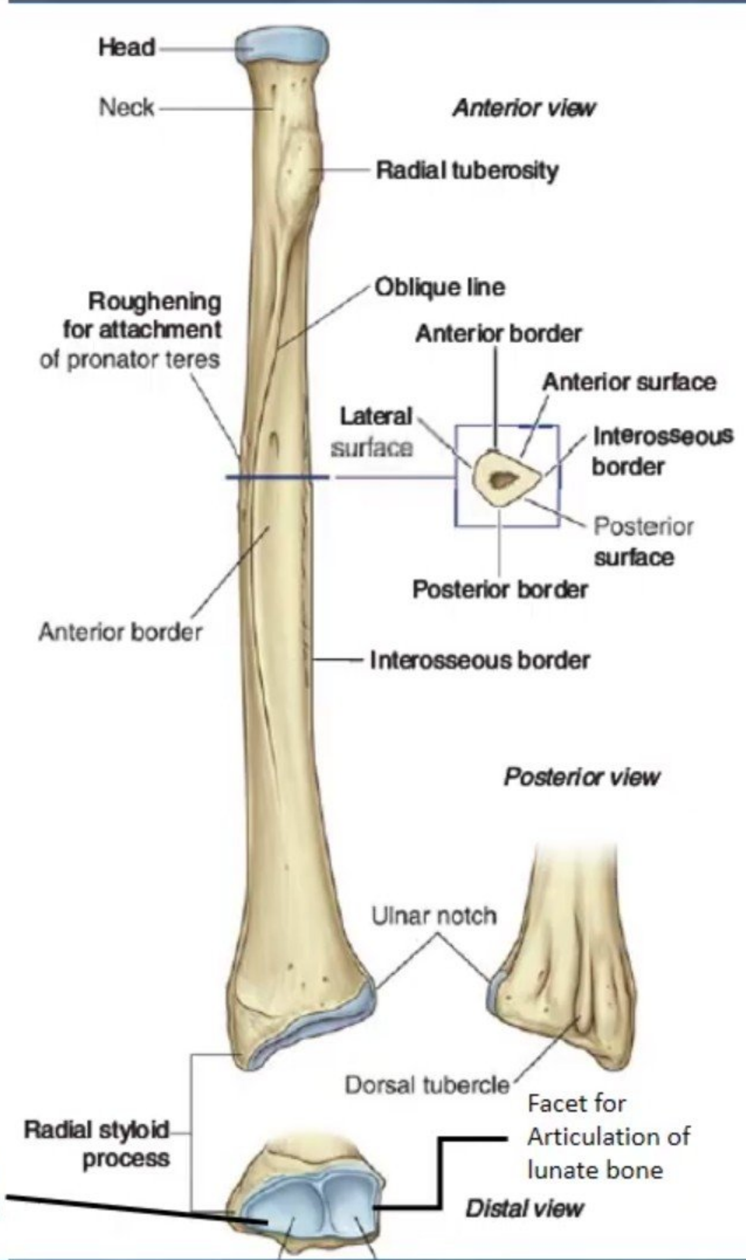
2.Radial Styloid process: extends from the lateral aspect.

3.Dorsal tubercle: projects dorsally.



Inferior view of distal ends of ulna and radius



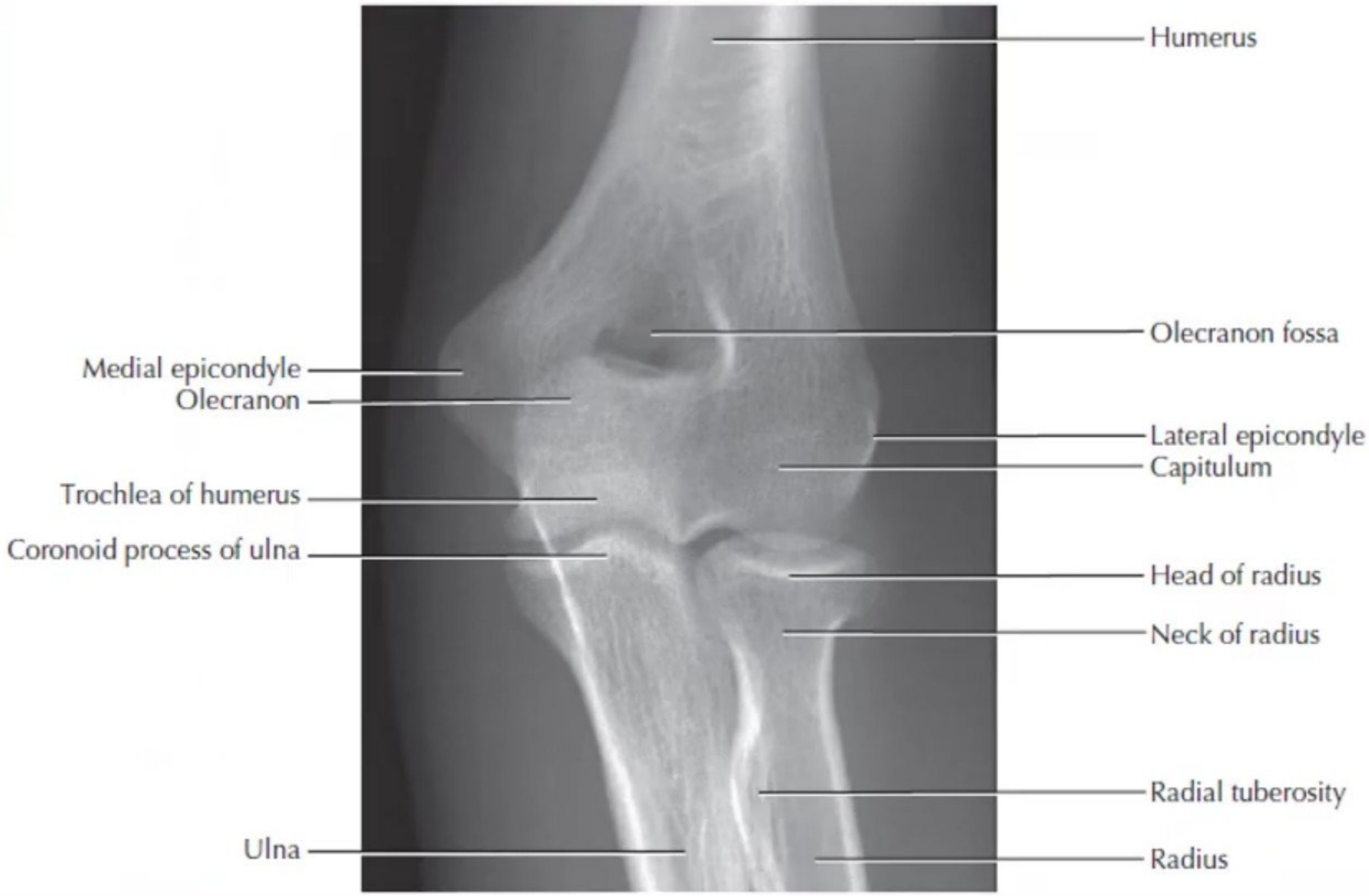


Normal X-ray Radiographic Appearances

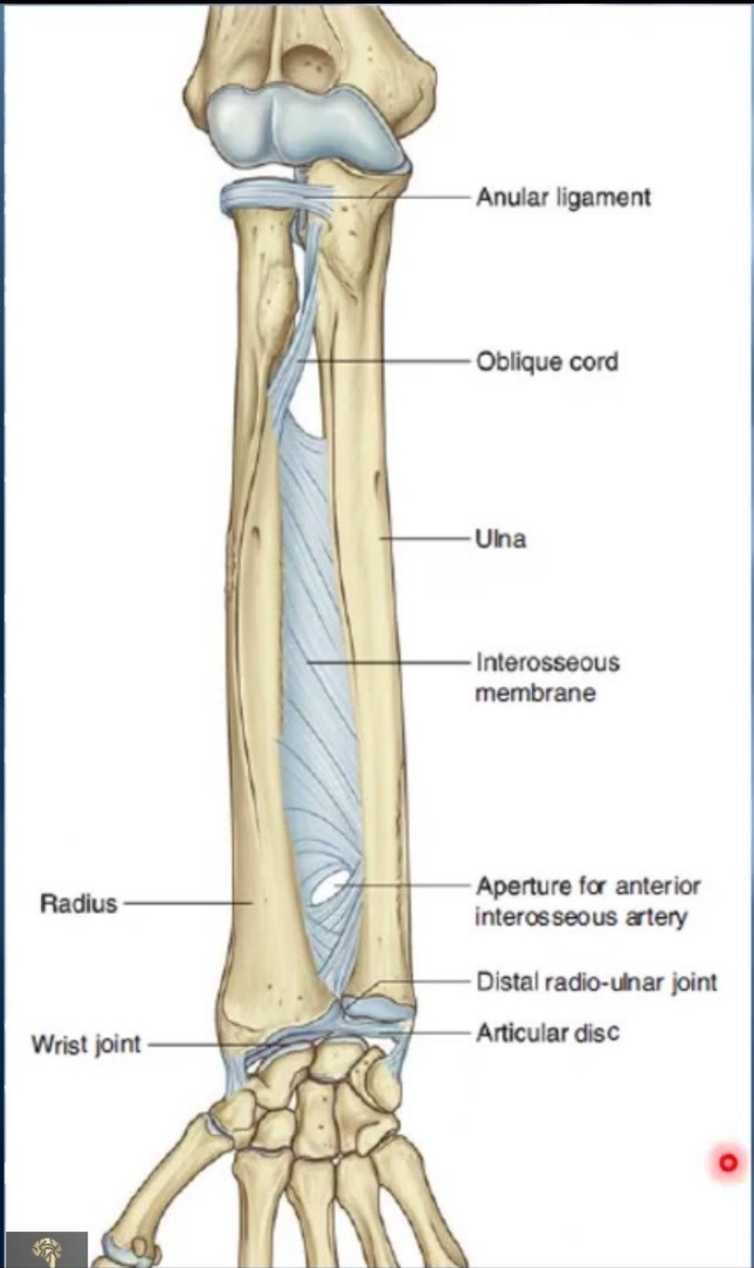
Anteroposterior radiograph of the elbow region in the adult.

In extension: posterior view

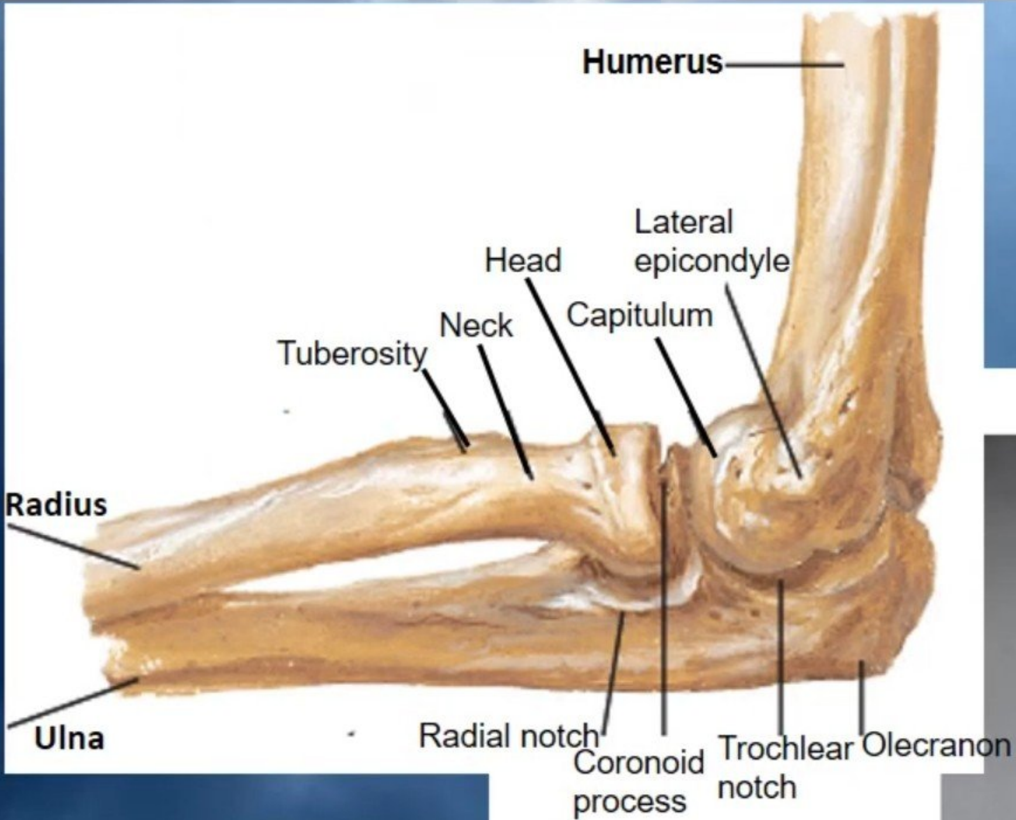
Anteroposterior radiograph



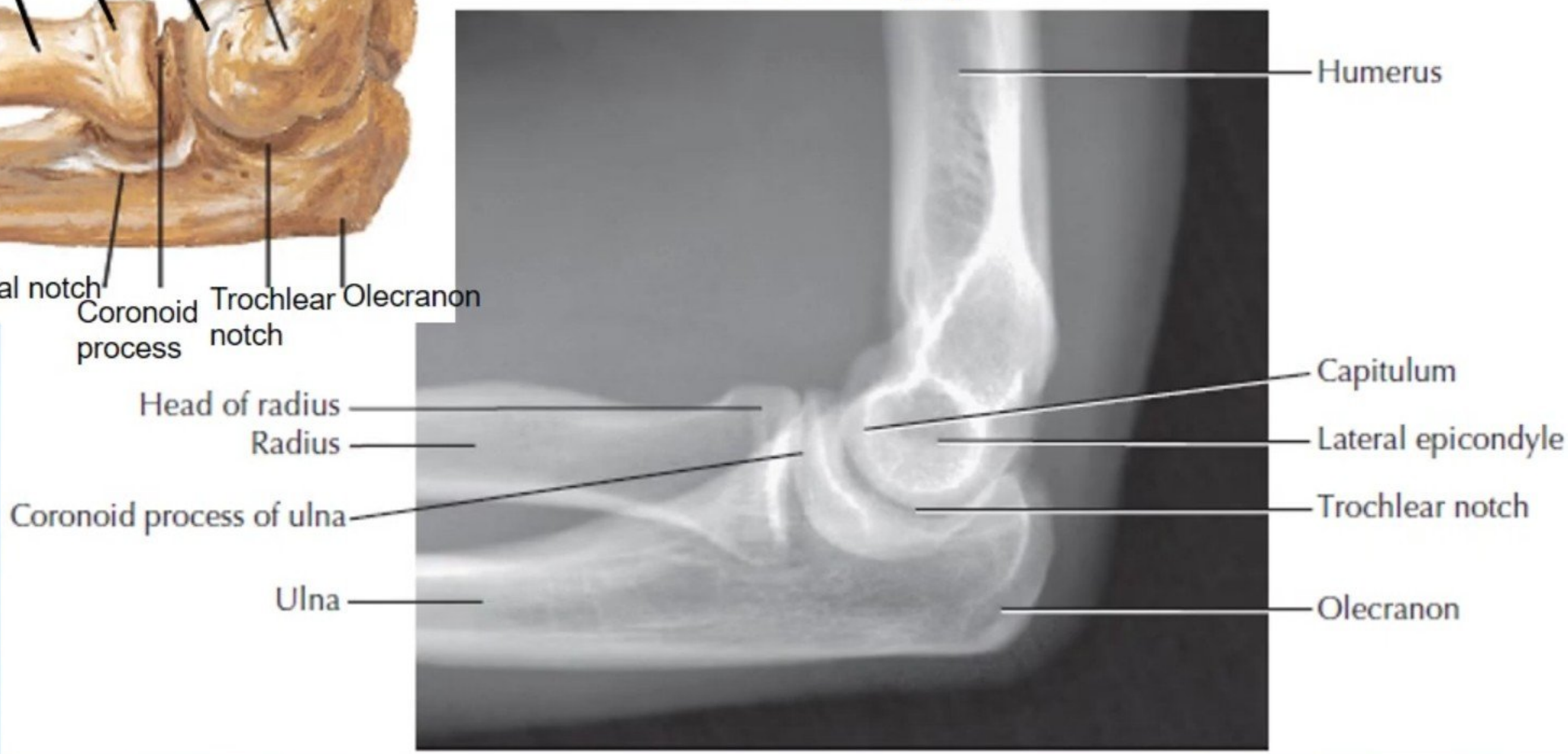
Normal X-ray Radiographic Appearances



Normal X-ray Radiographic Appearances

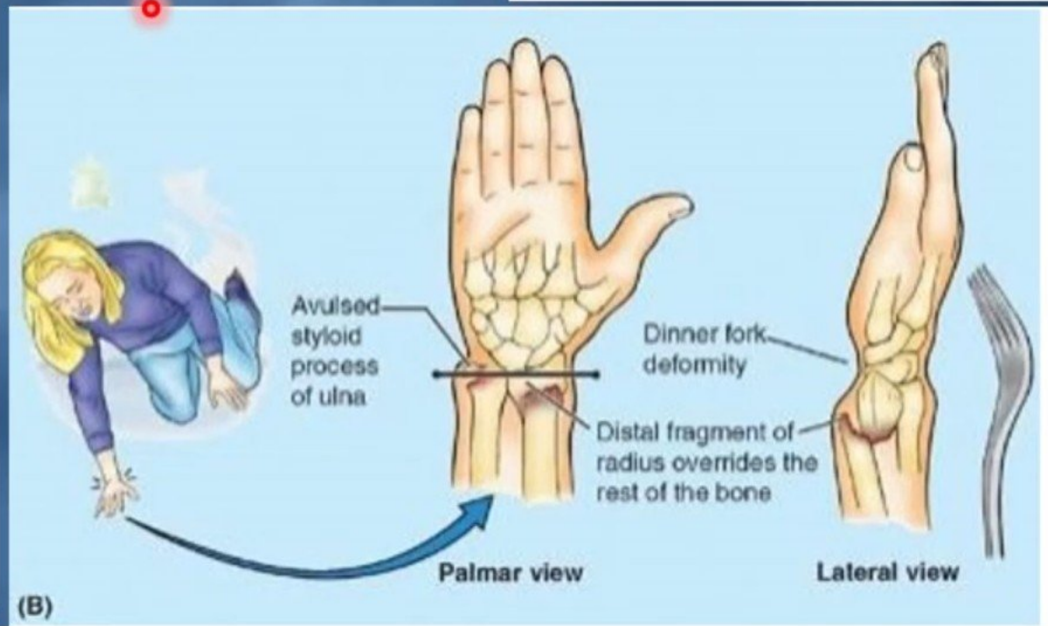


Lateral radiograph



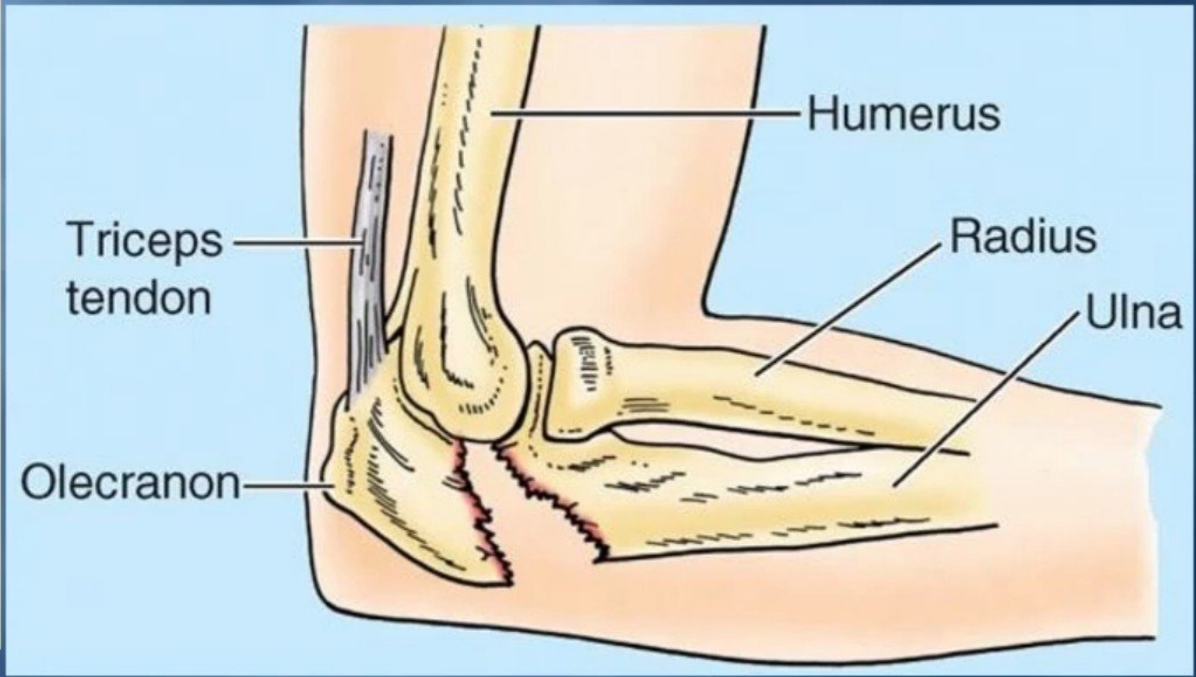
Fractures of radius & ulna

- *Because the radius & ulna are firmly bound by the* ***interosseous membrane****, a fracture of one bone is commonly associated with* ***dislocation of the nearest joint.***
- ***Colle's Fracture*** *(****fracture of the distal end of radius****)* ***is the most common fracture of the forearm.***
- *It is* ***more common in women after middle age because of osteoporosis.***
- ***It causes dinner fork deformity.***
- *It results from* ***forced dorsiflexion*** *of the hand as a result to ease a* ***fall by outstretching the upper limb.***
- *The typical history of the fracture includes* ***slipping.*** *Because of the rich blood supply to the distal end of the radius, bony union is usually good.*



fractured elbow

- **Fracture of the olecranon**, called a "fractured elbow" by laypersons.
- It is common because the olecranon is subcutaneous and protrusive.
- The typical mechanism of injury is a **fall on the elbow** combined with sudden powerful contraction of the **triceps brachii**.
- The **fractured olecranon** is pulled away by **the active** and **tonic contraction** of the triceps.
- Healing occurs **slowly.**

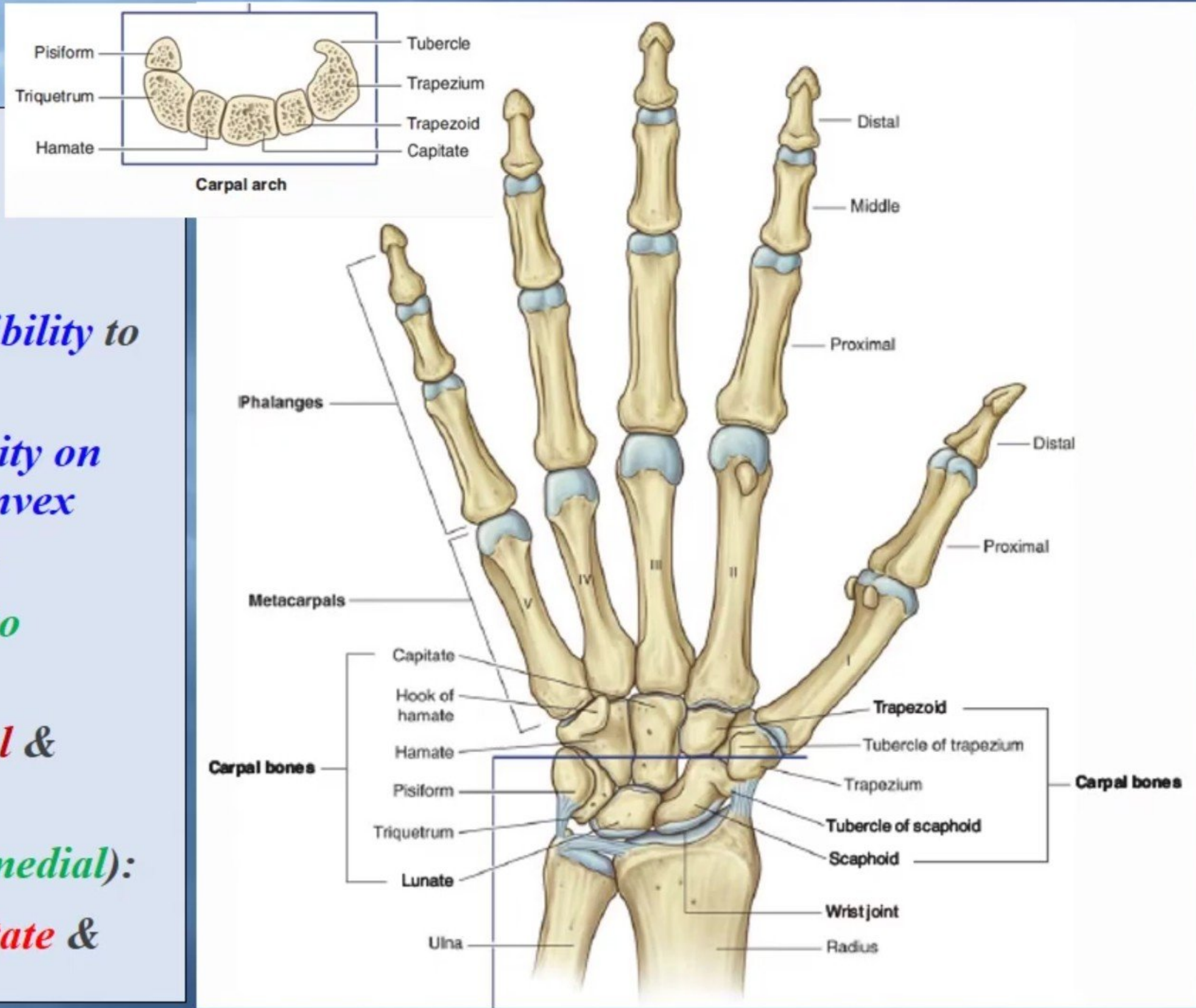


WRIST (CARPUS)

- *Composed of* ***Eight Carpal***

(short bones) arranged in ***two irregular rows****,* ***Four each****.*

- ***These Small bones give flexibility to the wrist.***
- ***The carpus presents Concavity on their Anterior surface & Convex from side to side Posteriorly.***
- ***Proximal row (from lateral to medial):***
- ***Scaphoid, Lunate, Triquetral & Pisiform bones.***
- ***Distal row (from lateral to medial):***
- ***Trapezium, Trapezoid, Capitate & Hamate.***



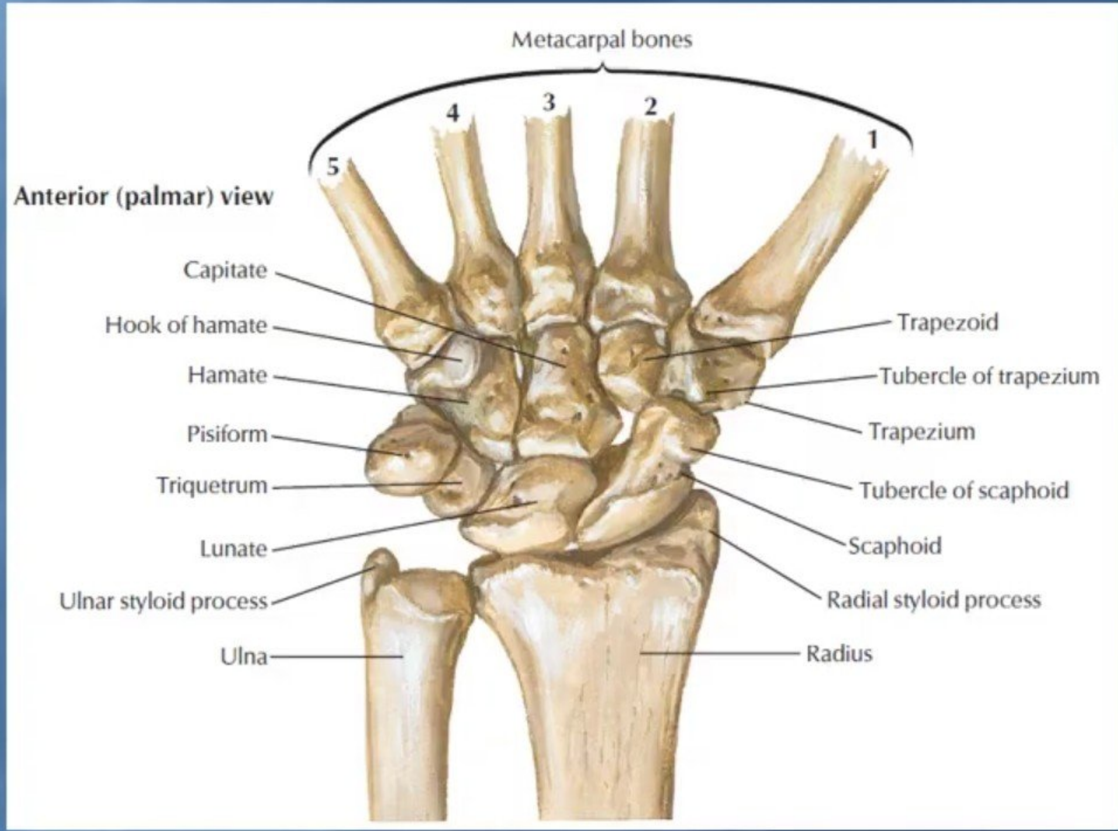
Carpal Bones

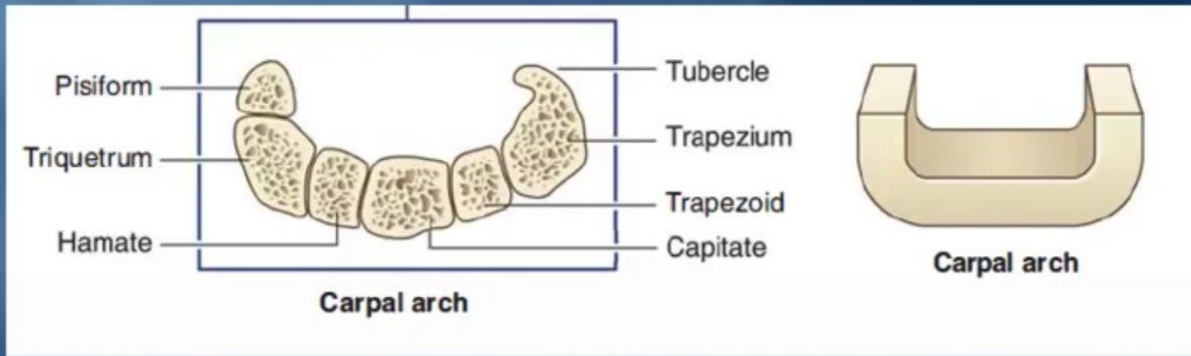
- **S**he – **S**caphoid.
- **L**ooks – **L**unate.
- **T**oo – **T**riquetral.
- **P**retty – **P**isiform.

From lateral to medial Proximal row.

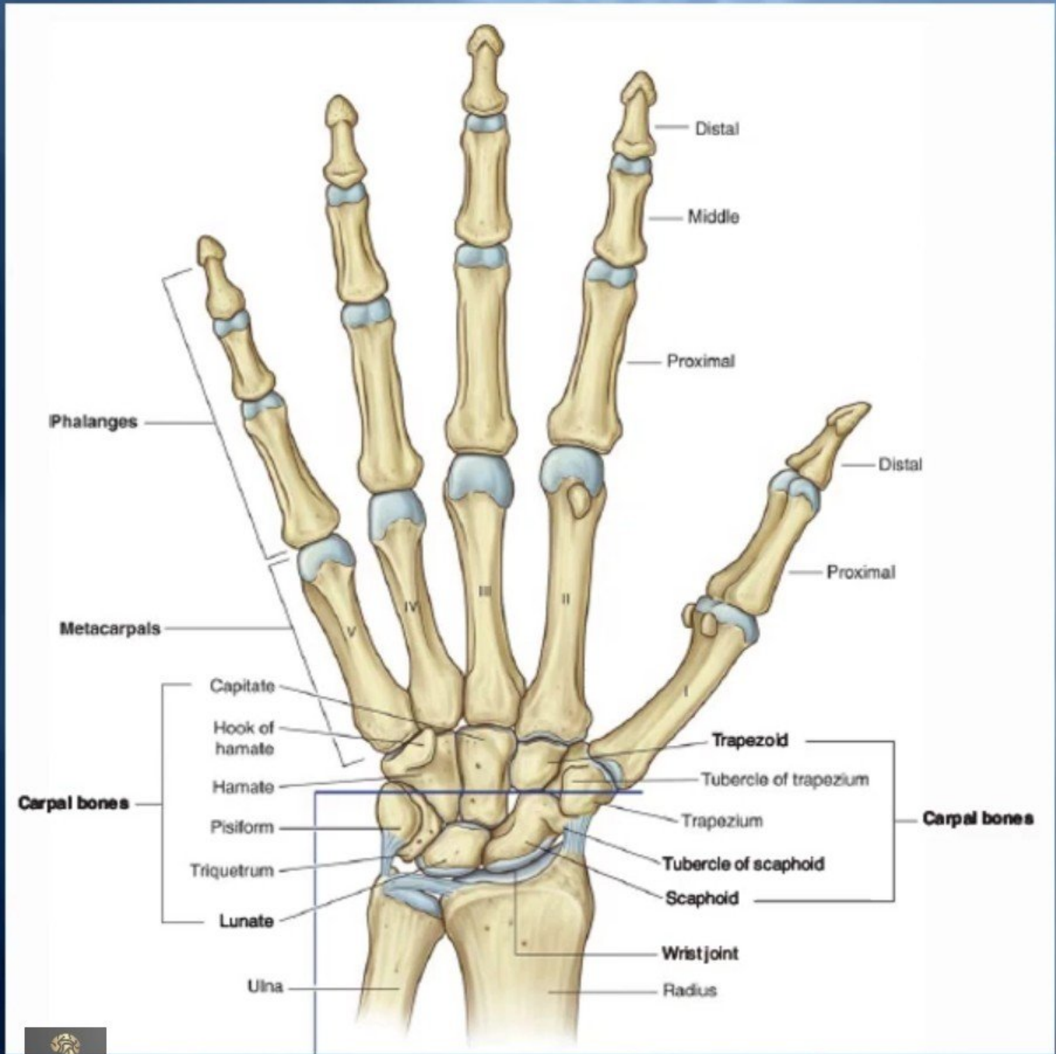
- **T**ry – **T**rapezium.
- **T**o – **T**rapezoid.
- **C**atch – **C**apitate.
- **H**er – **H**ammate.

From lateral to medial Distal row.

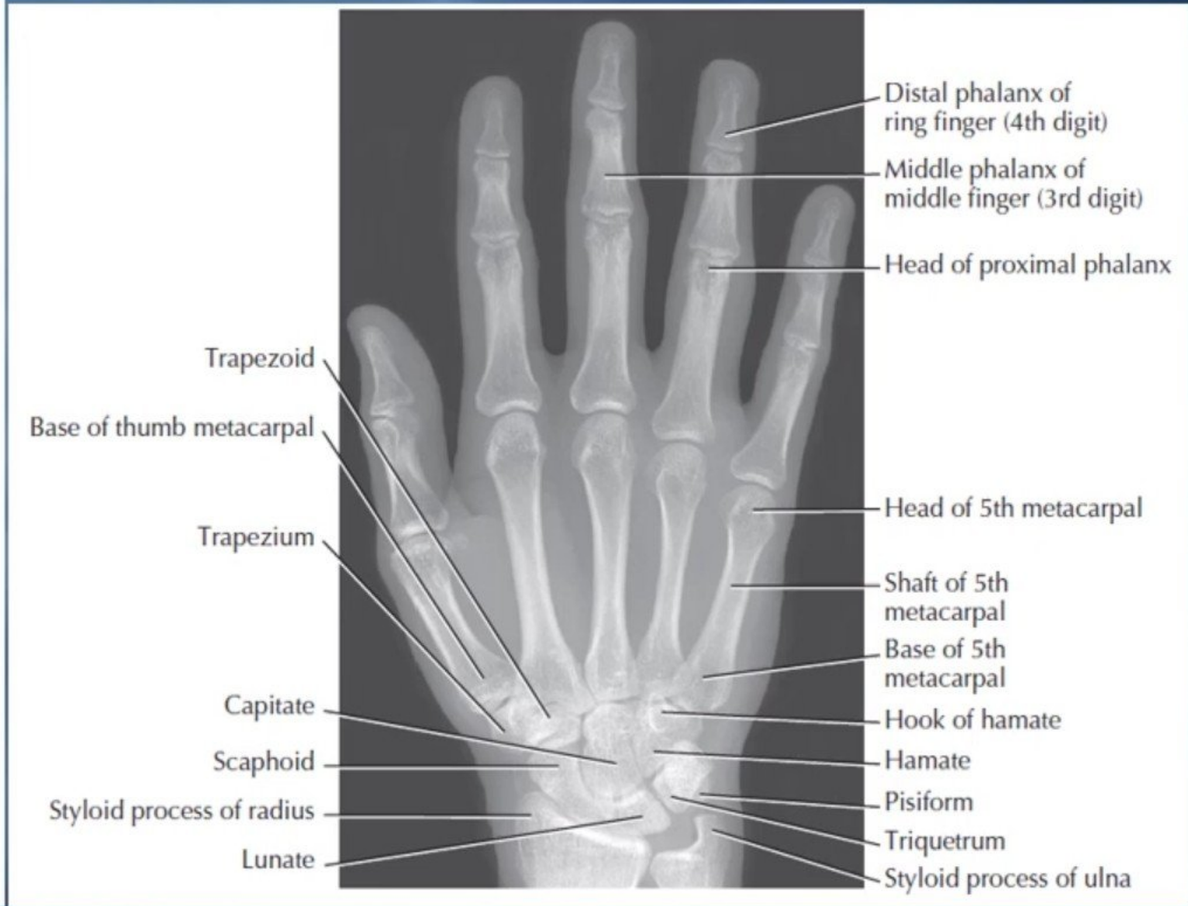




Normal X-ray Radiographic Appearances



Posteroanterior radiograph of an adult wrist and hand



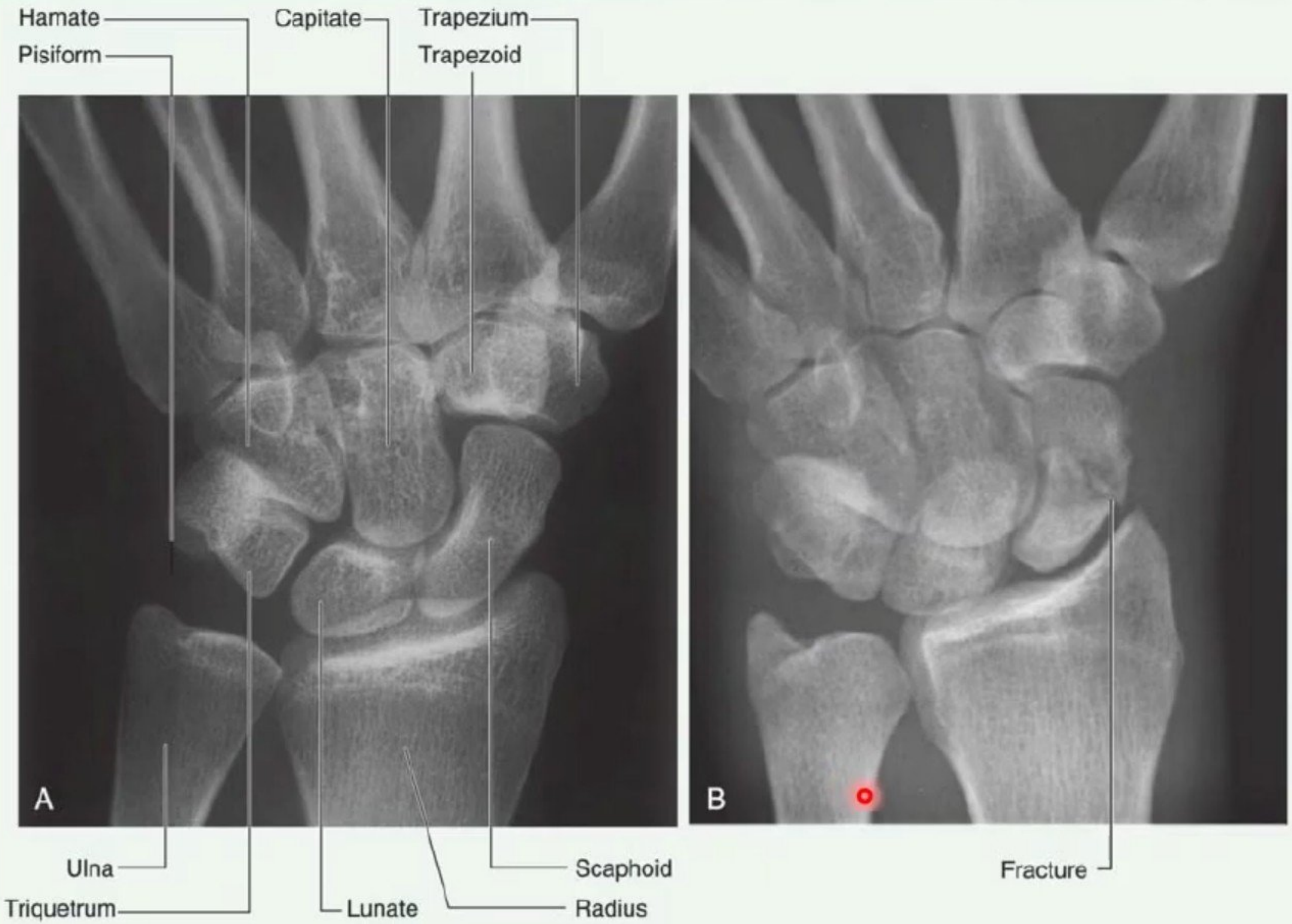
Scaphoid fracture

It is the most commonly fractured carpal bone and it is the most common injury of the wrist.

It is the result of a **fall into the palm** when the hand is **abducted**.

Pain occurs **along the lateral side of the wrist** especially during dorsiflexion and abduction of the hand.

Union of the bone may take several months because of poor blood supply to the proximal part of the scaphoid.



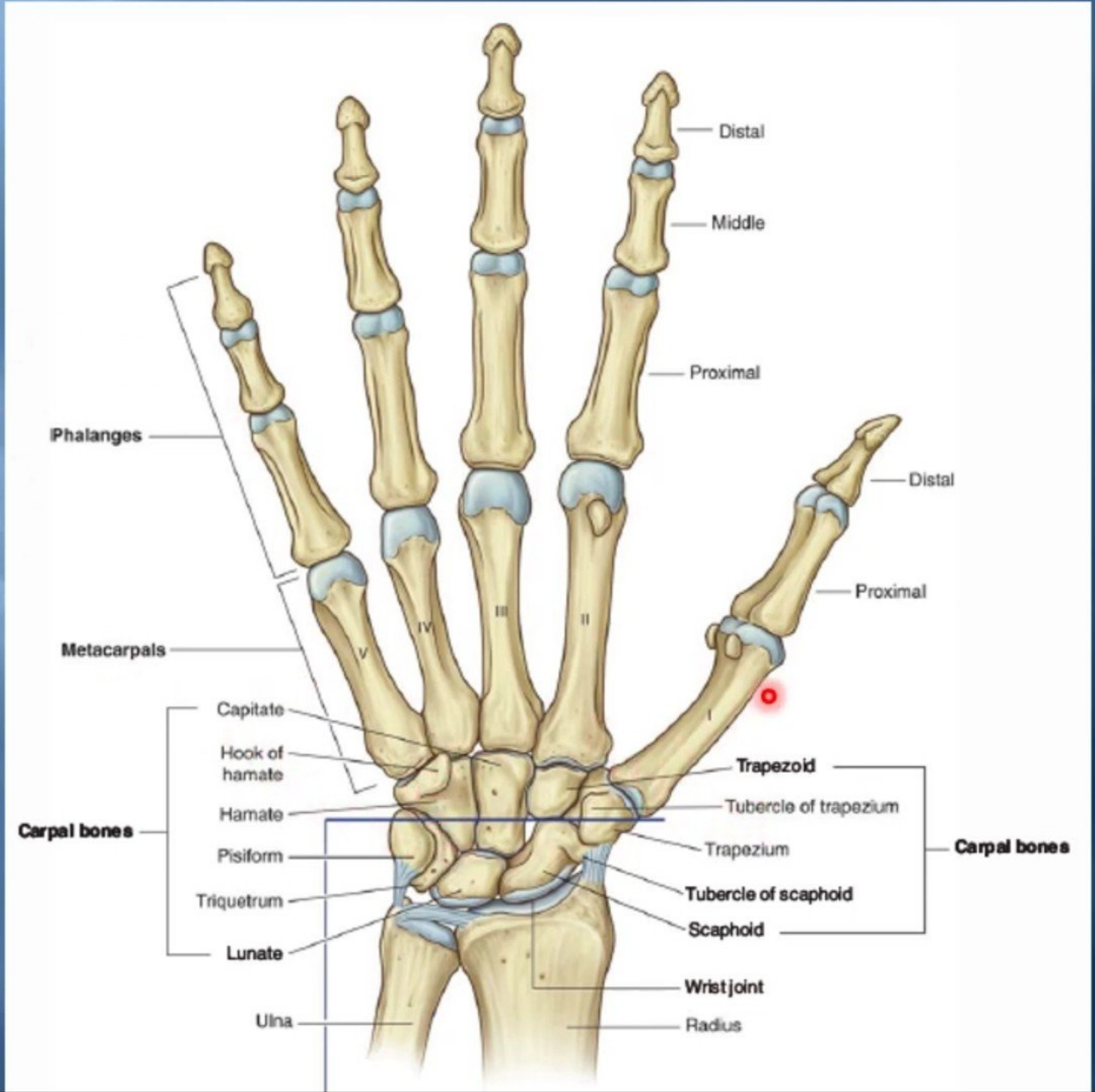
Important notes

Scaphoid: is the commonly fractured = avascular necrosis
Lunate: most commonly dislocated = anteriorly .
Triquetrum: pisiform lie anterior to it .
Pisiform: sesamoid bone= flexor carpi unlnaris / Ossify last.
Trapezium: groove for flexor carpi radialis ant./ radial artery post.
Capitate: largest ! Ossify first
Hamate: hook deep branch of ulnar nerve

* All ossify in cartilage by age of 7 years

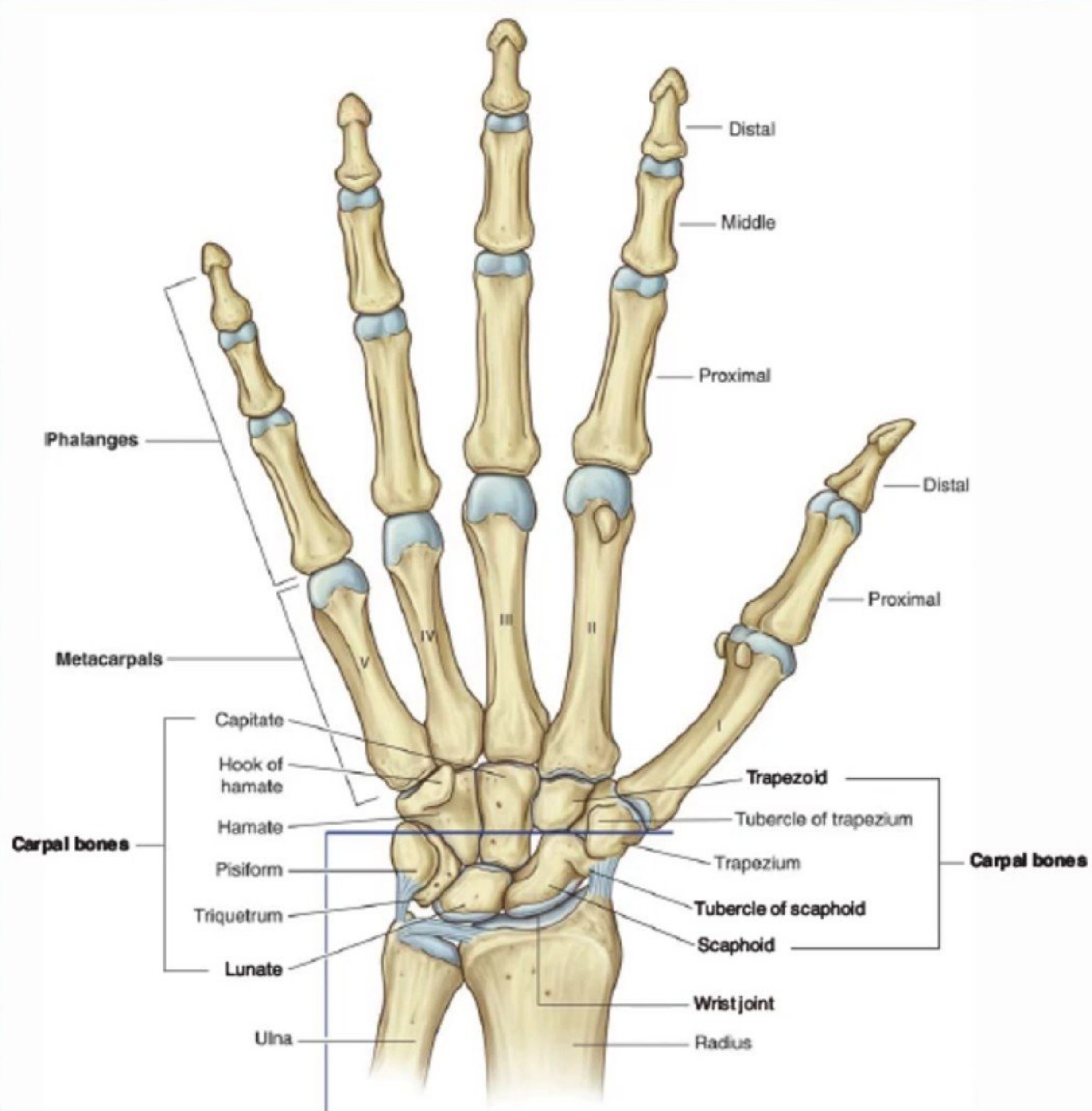
Metacarpus

- It is the skeleton of the hand between the carpus and phalanges.
- It is composed of Five Metacarpal bones, each has a Base, Shaft, and a Head.
- They are numbered 1-5 from the thumb.
- The distal ends (Heads) articulate with the proximal phalanges to form the Knuckles of the fist.
- The Bases of the metacarpals articulate with the carpal bones.
- The 1st metacarpal is the shortest and most mobile.



Digits Phalanges

- *Each digit has **Three Phalanges***
- ***Except** the Thumb which has only **Two***
- ***Each phalanx has a Base Proximally, a Head distally and a Body between the base and the head.***
- ***The proximal phalanx is the largest.***
- ***The middle ones are intermediate in size.***
- ***The distal ones are the smallest, its distal ends are flattened and expanded distally to form the nail beds.***




Thanks